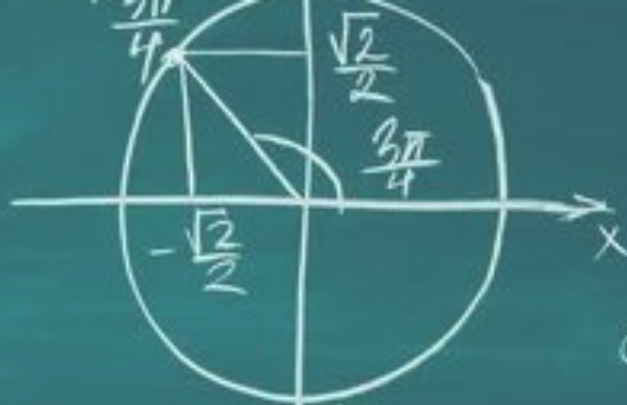
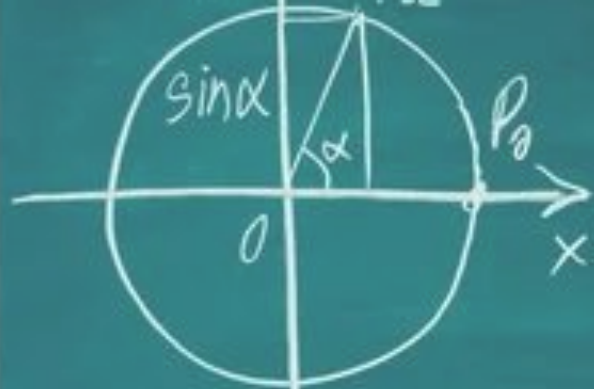




$$\operatorname{tg} \frac{3\pi}{4} = -1$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \operatorname{tg} \frac{3\pi}{4} = -1$$

$$D(\sin) = D(\cos) = 0 \quad E(\sin) = E(\cos) = [1; 1]$$

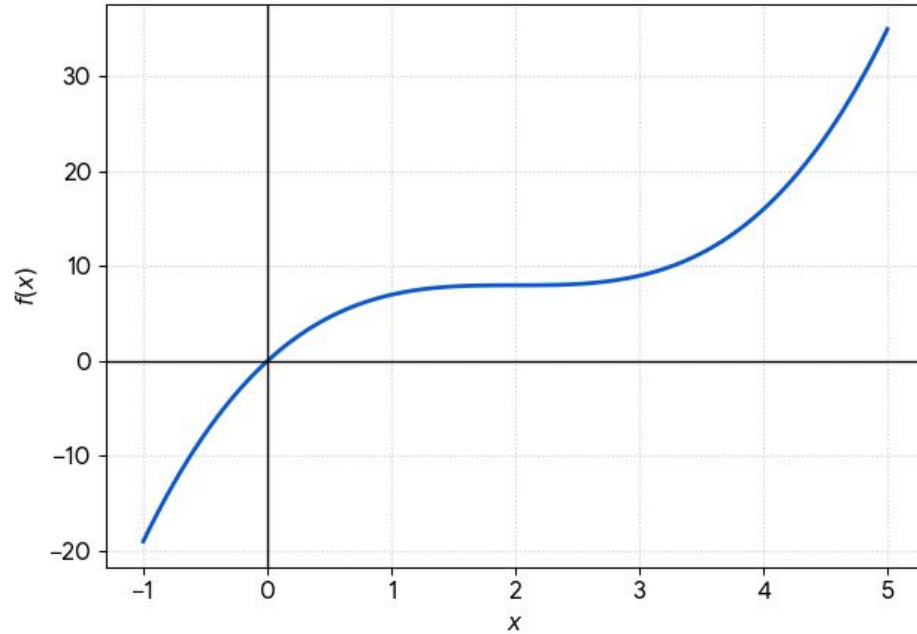
First year Aerospace Engineering

Functions exercises

Marcos Carrera Valle

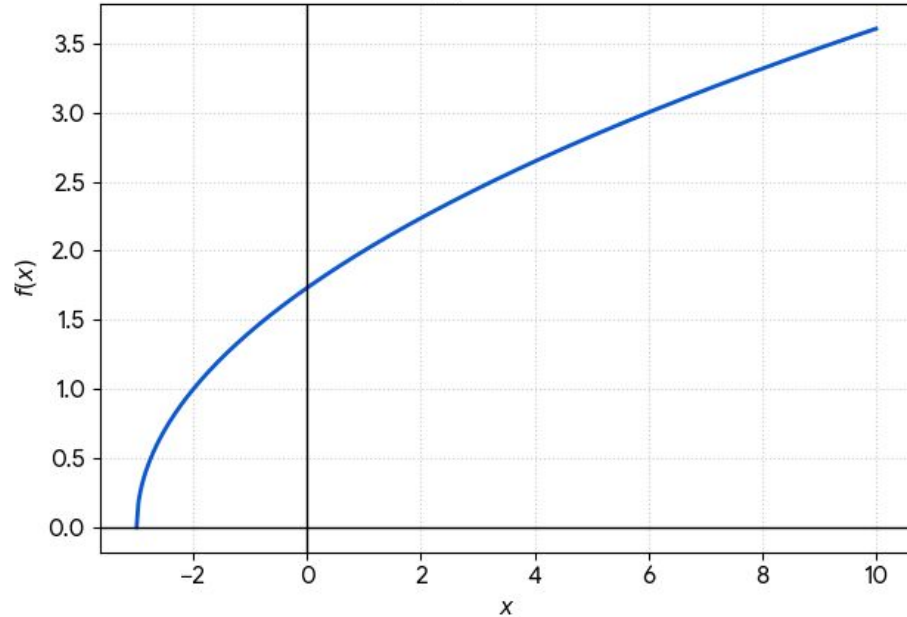
1) Which function does this curve represent?

- A. $f(x) = (x+2)^3$
- B. $f(x) = x^3 - 2$
- C. $f(x) = (x-2)^3 + 8$
- D. $f(x) = 1/2(x-2)^3$



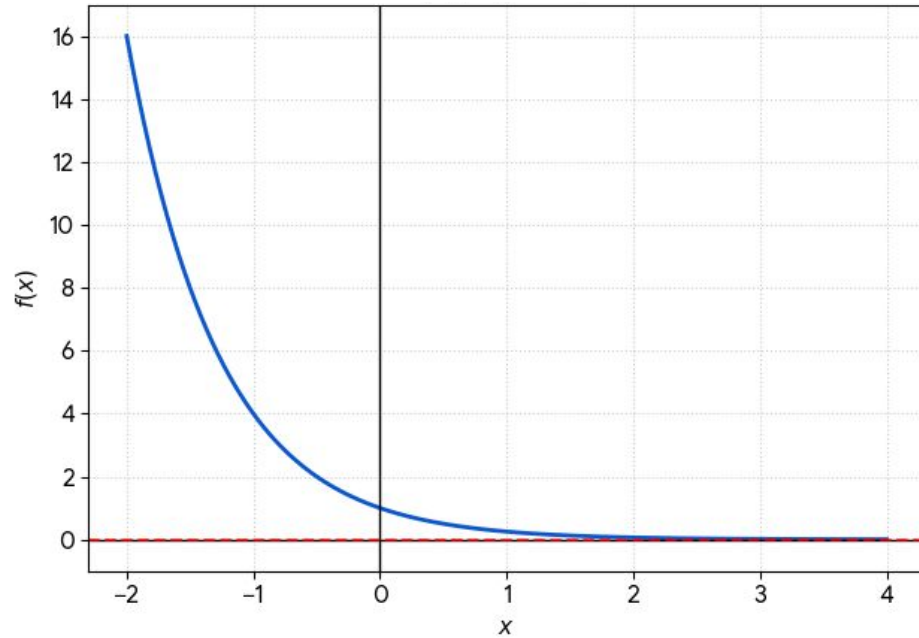
2) Which function does this curve represent?

- A. $f(x) = \sqrt{x} - 3$
- B. $f(x) = \sqrt{x+3}$
- C. $f(x) = \sqrt{x-3}$
- D. $f(x) = 2\sqrt{x+3}$



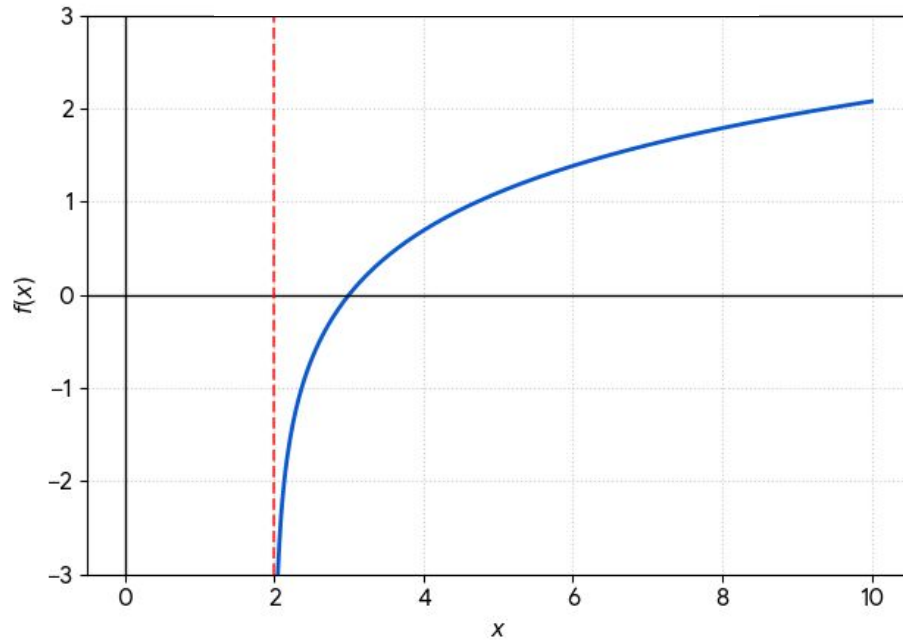
3) Which function does this curve represent?

- A. $f(x) = 4^x$
- B. $f(x) = e^{(x-1)}$
- C. $f(x) = (0.25)^x$
- D. $f(x) = -4^x$



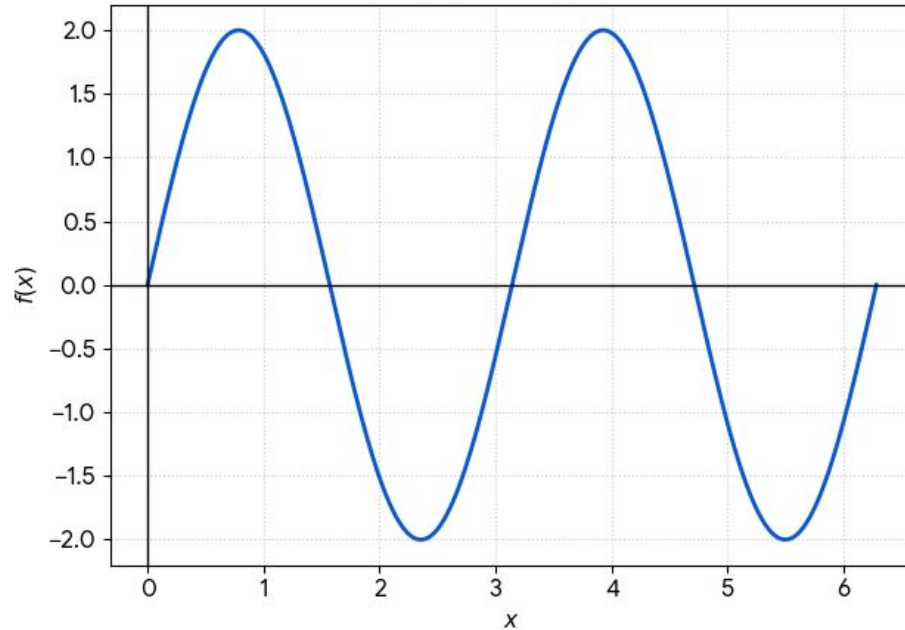
4) Which function does this curve represent?

- A. $f(x) = \ln(x) + 2$
- B. $f(x) = \ln(x+2)$
- C. $f(x) = \ln(x-2)$
- D. $f(x) = 2 \cdot \ln(x)$



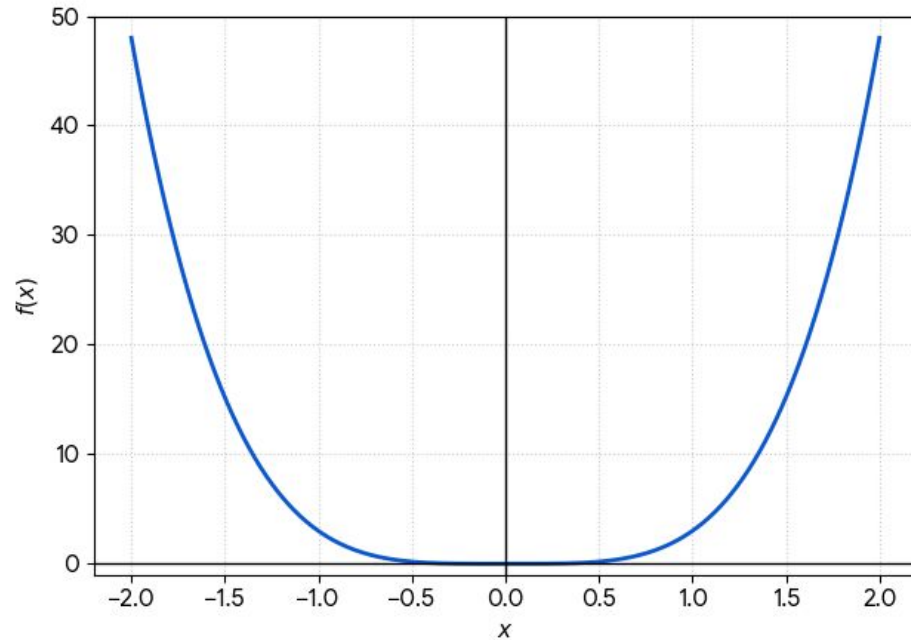
5) Which function does this curve represent?

- A. $f(x) = 2\sin(x)$
- B. $f(x) = \sin(2x)$
- C. $f(x) = 2\sin(2x)$
- D. $f(x) = 1/2\sin(\pi x)$



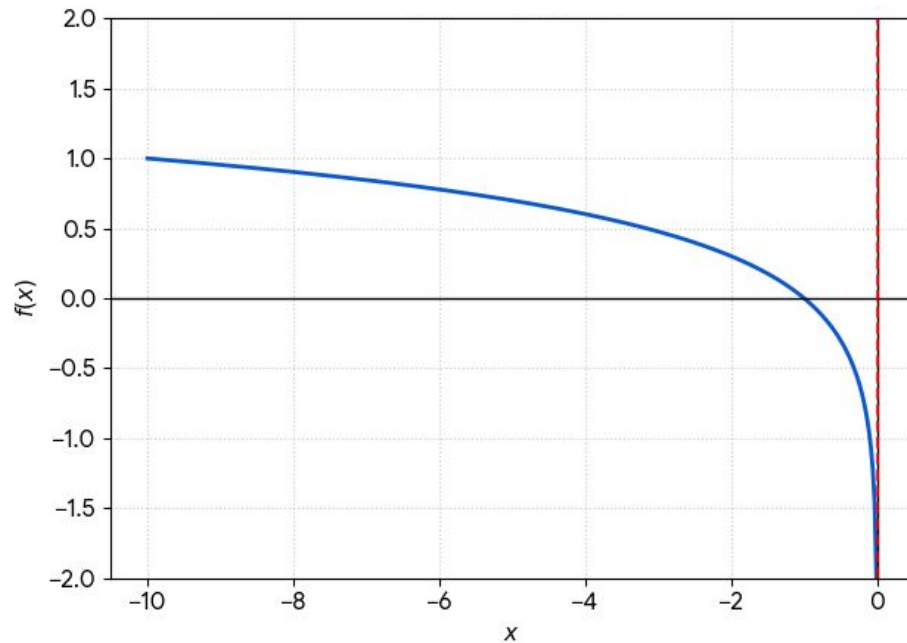
6) Which function does this curve represent?

- A. $f(x) = 3x^2$
- B. $f(x) = x^4 + 2$
- C. $f(x) = 3x^4$
- D. $f(x) = x^6 + 3$



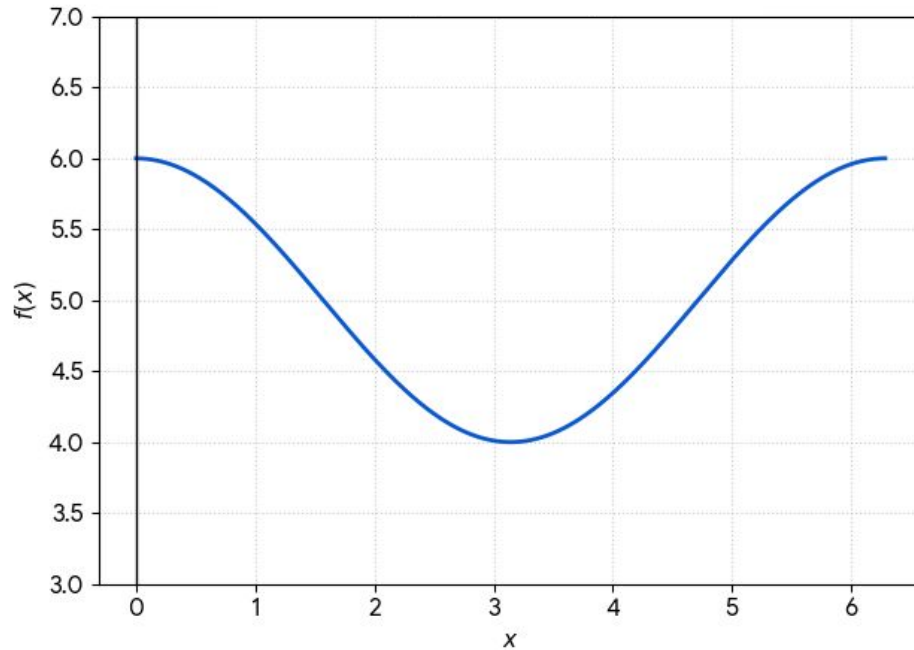
7) Which function does this curve represent?

- A. $f(x) = \log_{10}(x)$
- B. $f(x) = -\log_{10}(x)$
- C. $f(x) = \log_{10}(-x)$
- D. $f(x) = \log_{10}(x-10)$



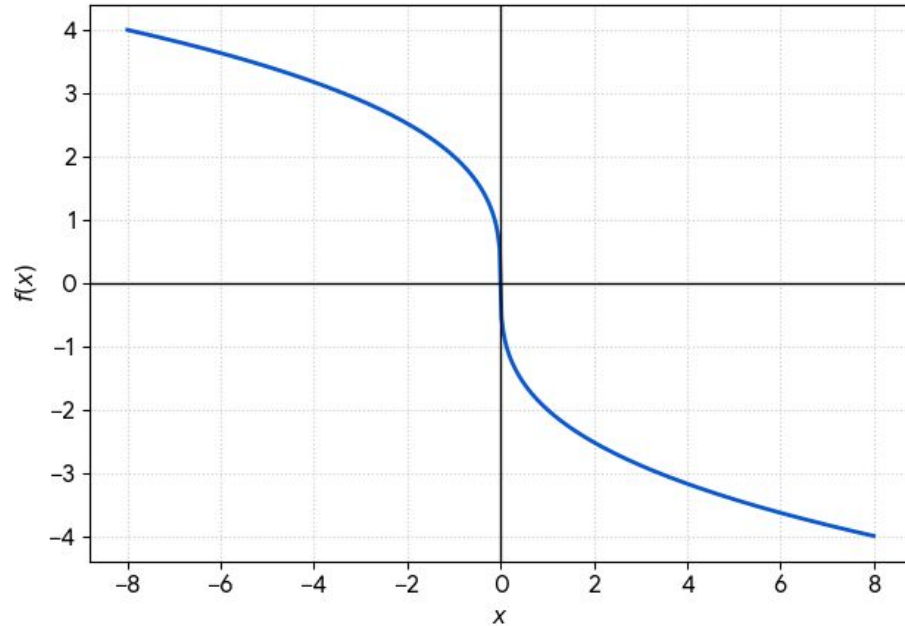
8) Which function does this curve represent?

- A. $f(x) = 5 + \cos(x)$
- B. $f(x) = 6\cos(x)$
- C. $f(x) = \cos(x) + 4$
- D. $f(x) = 2\cos(x) + 4$



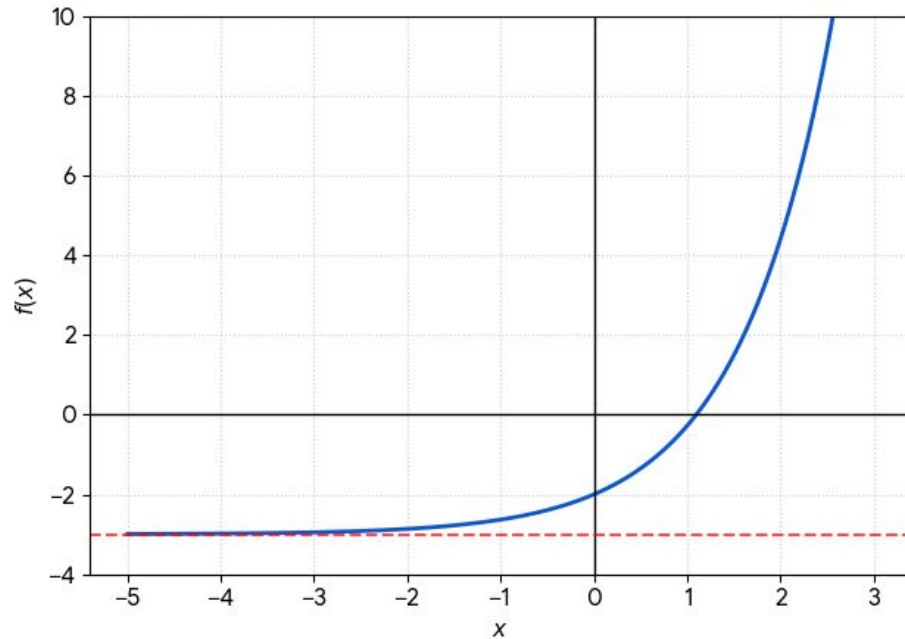
9) Which function does this curve represent?

- A. $f(x) = -2 \cdot \text{cbrt}(x)$
- B. $f(x) = \text{cbrt}(x-2)$
- C. $f(x) = \text{cbrt}(-2x)$
- D. $f(x) = \text{cbrt}(x) - 2$



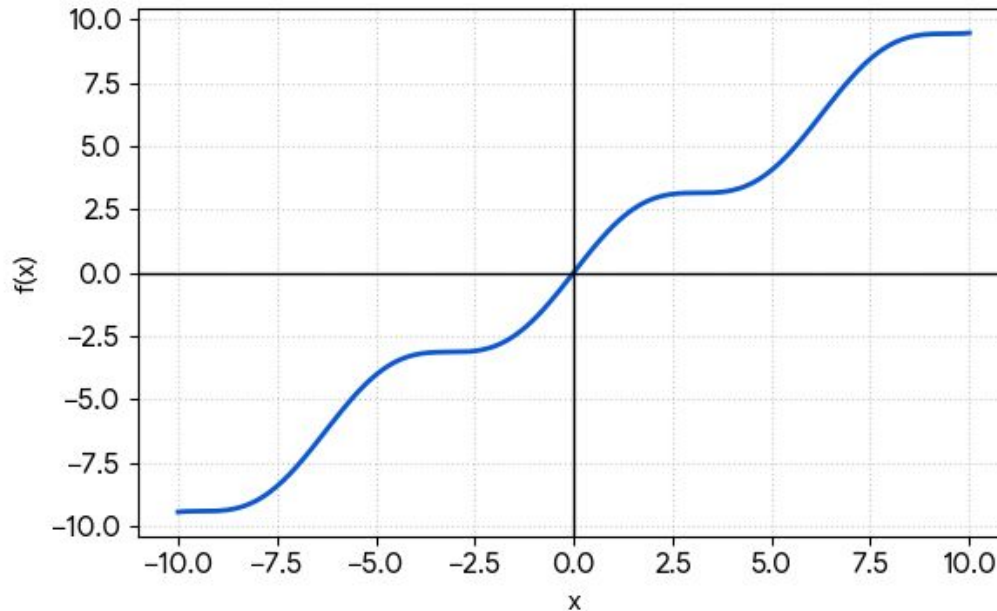
10) Which function does this curve represent?

- A. $f(x) = e^x + 3$
- B. $f(x) = e^{(x+3)}$
- C. $f(x) = e^x - 3$
- D. $f(x) = 3 \cdot e^x - 3$



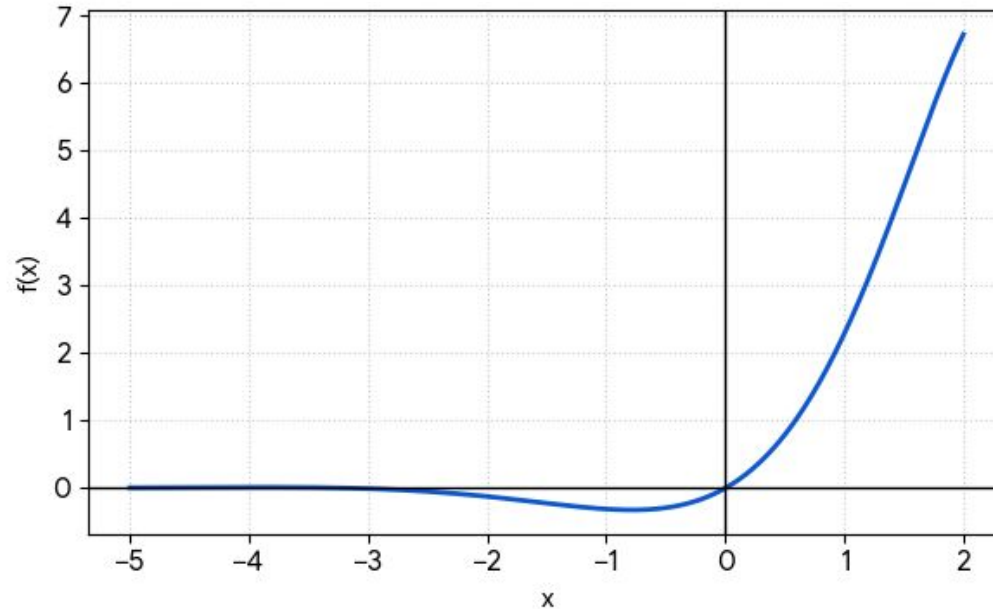
11) Which function does this curve represent?

- A. $f(x) = x + \sin(x)$
- B. $f(x) = x * \sin(x)$
- C. $f(x) = x + \cos(x)$
- D. $f(x) = x^2 + \sin(x)$



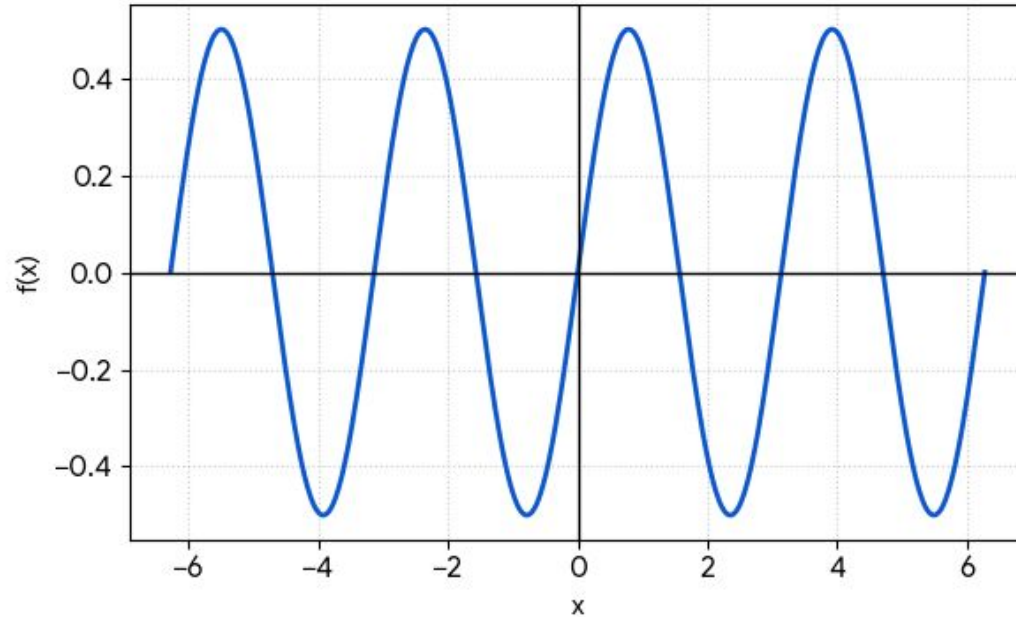
12) Which function does this curve represent?

- A. $f(x) = e^x + \sin(x)$
- B. $f(x) = e^x * \sin(x)$
- C. $f(x) = e^{-x} * \sin(x)$
- D. $f(x) = \ln(x) * \sin(x)$



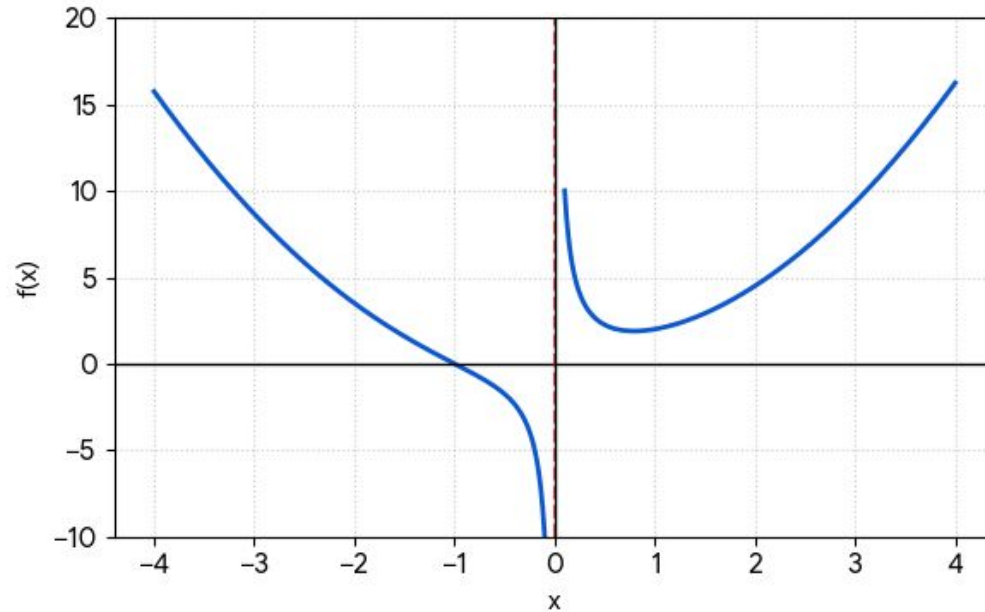
13) Which function does this curve represent?

- A. $f(x) = \sin(x) + \cos(x)$
- B. $f(x) = \sin(x) * \cos(x)$
- C. $f(x) = \sin(x) / \cos(x)$
- D. $f(x) = \sin(x) + x^2$



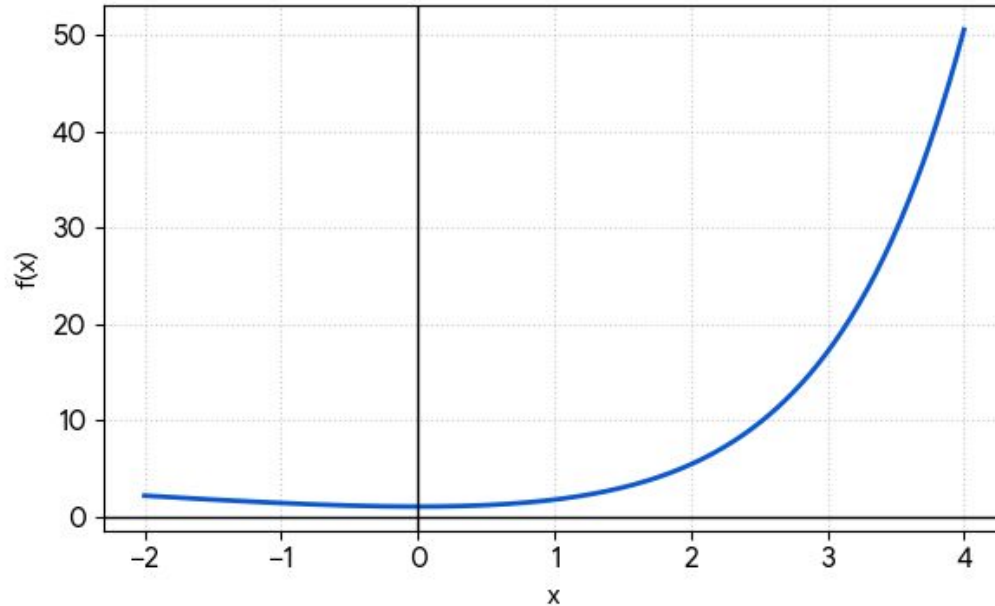
14) Which function does this curve represent?

- A. $f(x) = x^2 + \ln(x)$
- B. $f(x) = x^2 - e^x$
- C. $f(x) = x^2 + 1/x$
- D. $f(x) = x^2 * \ln(x)$



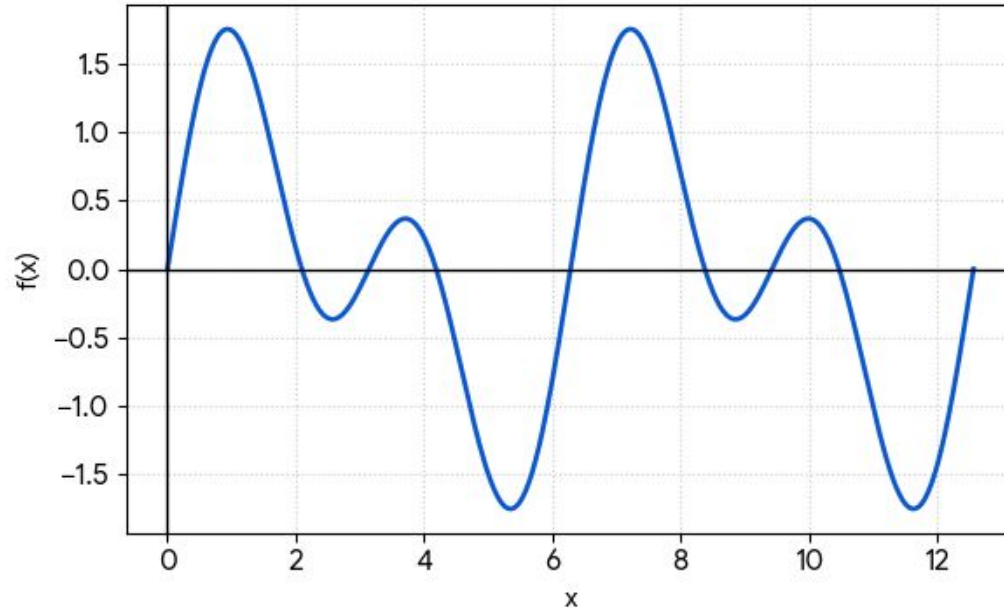
15) Which function does this curve represent?

- A. $f(x) = e^x + x$
- B. $f(x) = e^x - x$
- C. $f(x) = e^{-x} + x$
- D. $f(x) = e^x \cdot x$



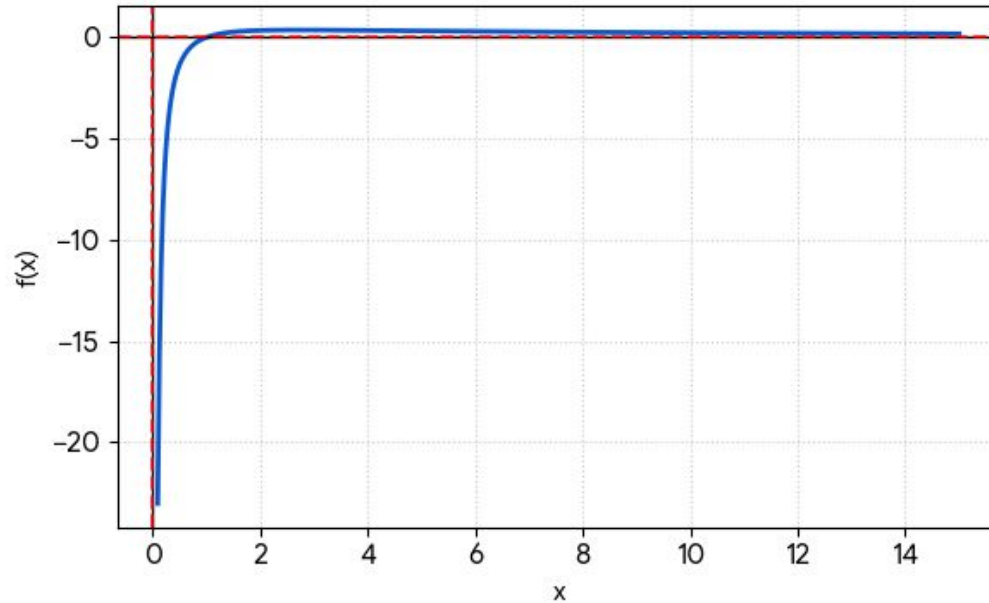
16) Which function does this curve represent?

- A. $f(x) = \sin(x) + \sin(2x)$
- B. $f(x) = \sin(x) + \cos(x)$
- C. $f(x) = \sin(x) * \sin(2x)$
- D. $f(x) = \text{abs}(\sin(x))$



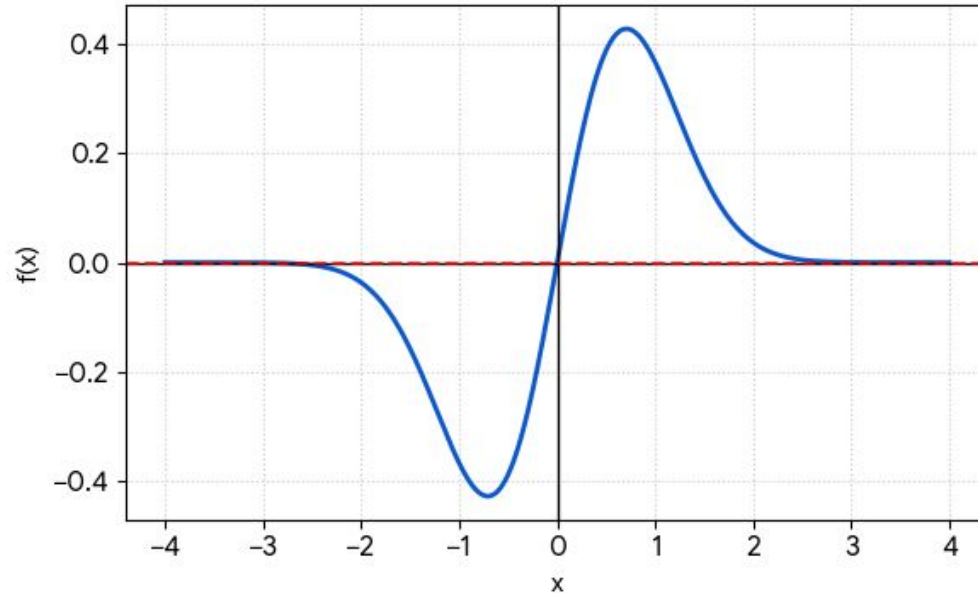
17) Which function does this curve represent?

- A. $f(x) = \ln(x) + x$
- B. $f(x) = \ln(x) - x$
- C. $f(x) = \ln(x) / x$
- D. $f(x) = \ln(x^2)$



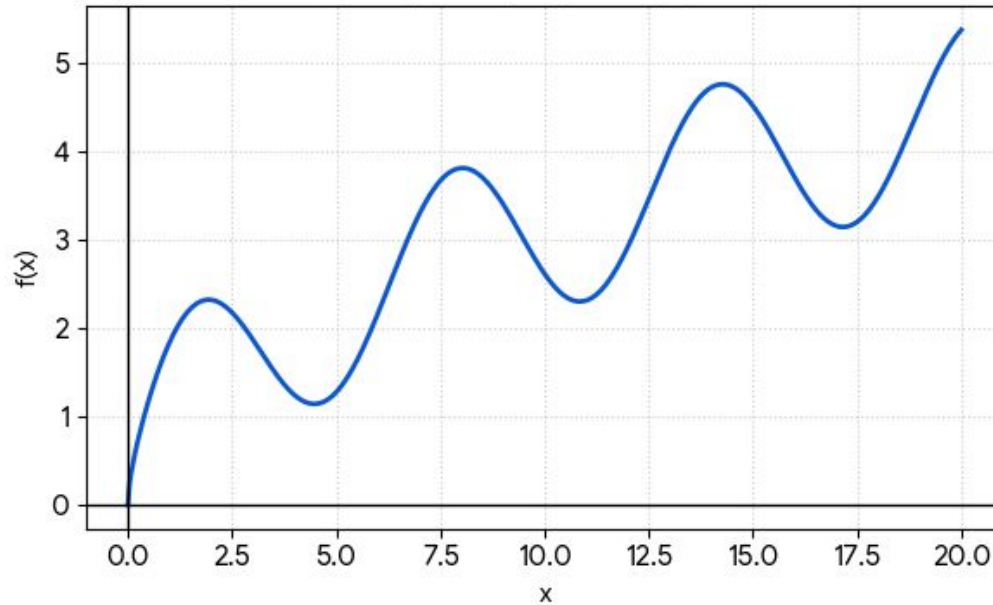
18) Which function does this curve represent?

- A. $f(x) = x * e^{(-x^2)}$
- B. $f(x) = x + e^{(-x)}$
- C. $f(x) = x^2 * e^x$
- D. $f(x) = e^{(-x)} * \cos(x)$



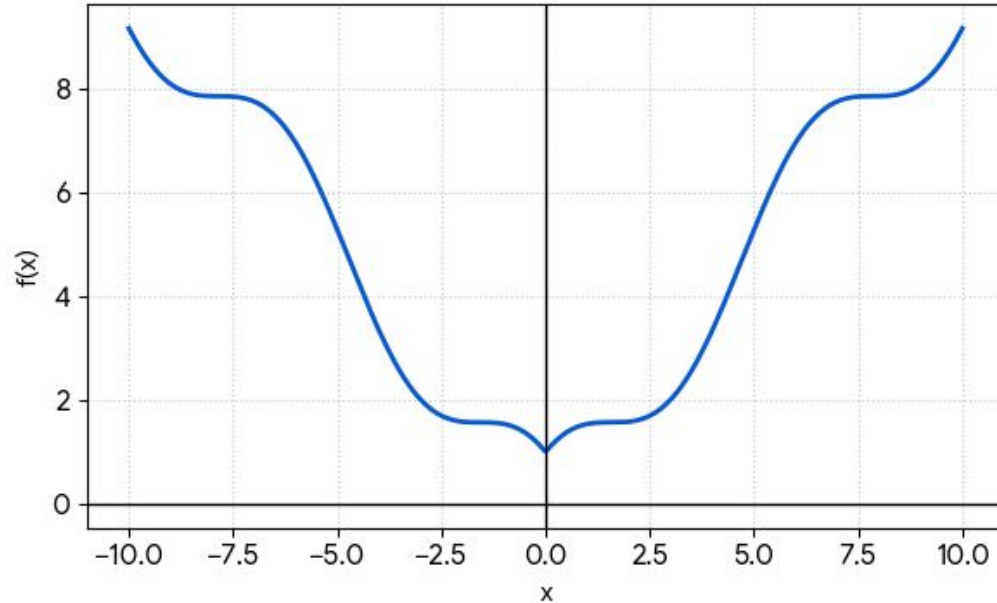
19) Which function does this curve represent?

- A. $f(x) = \sqrt{x} + \sin(x)$
- B. $f(x) = \sqrt{x} * \cos(x)$
- C. $f(x) = 1/x + \sin(x)$
- D. $f(x) = x^{1/3} + x$

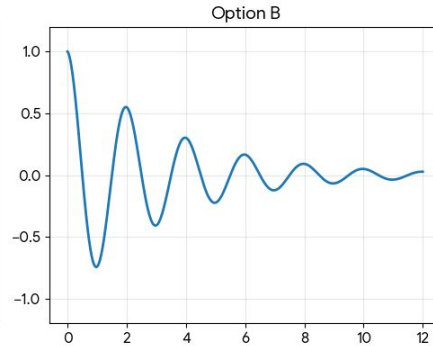
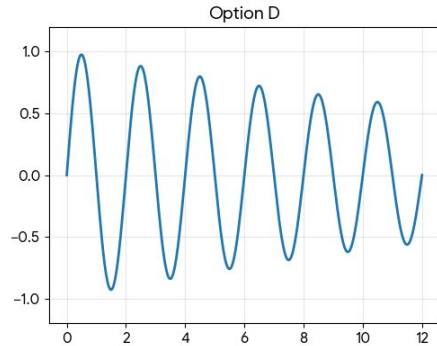
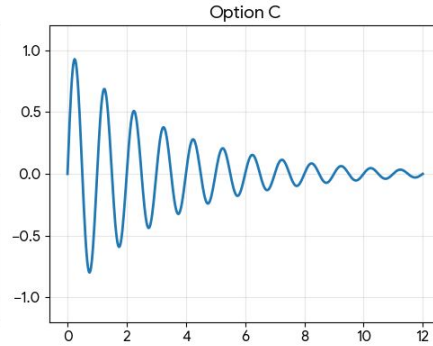
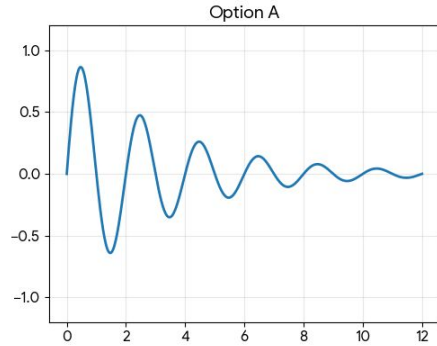


20) Which function does this curve represent?

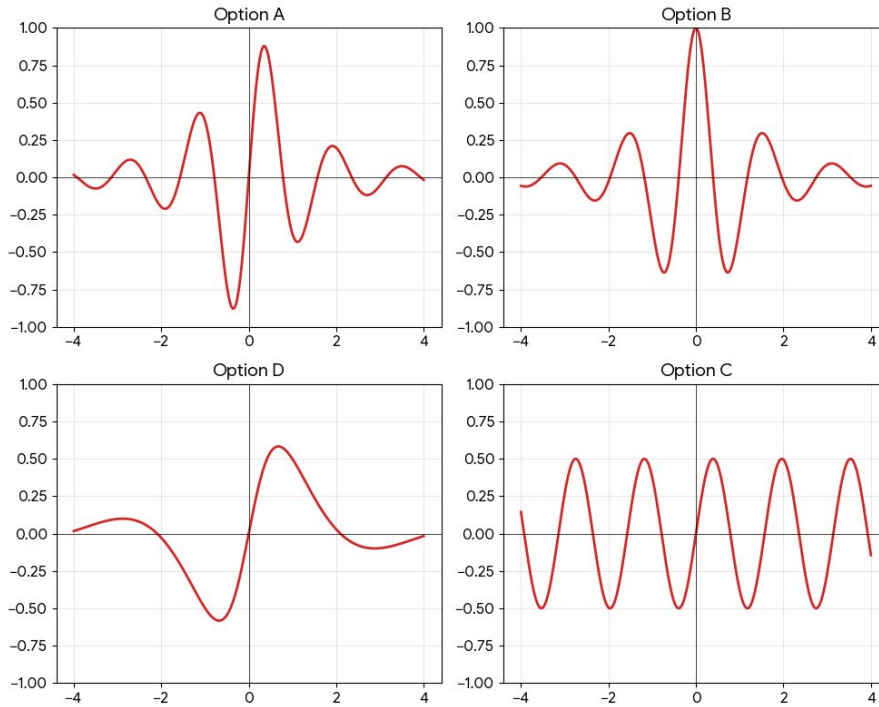
- A. $f(x) = \text{abs}(x) + \cos(x)$
- B. $f(x) = x^2 + \cos(x)$
- C. $f(x) = \text{abs}(x) * \sin(x)$
- D. $f(x) = 1/x + \cos(x)$



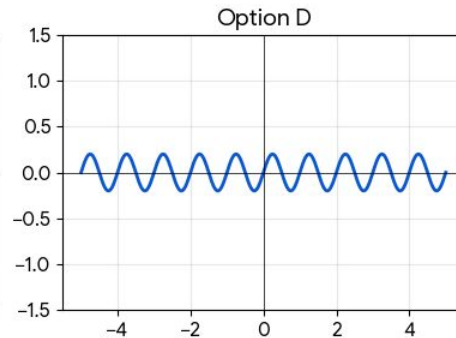
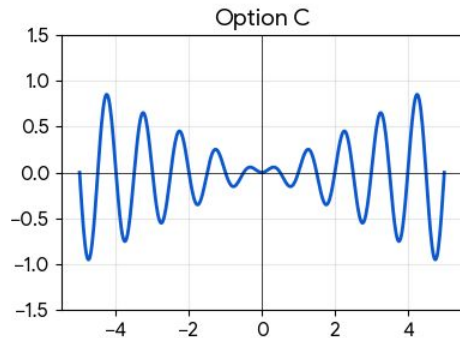
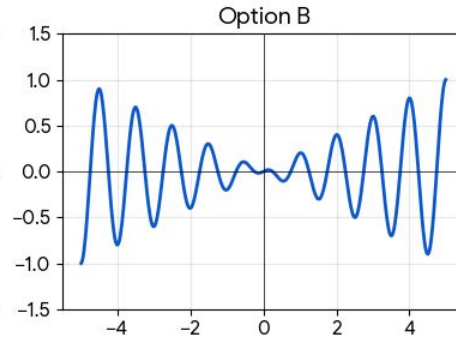
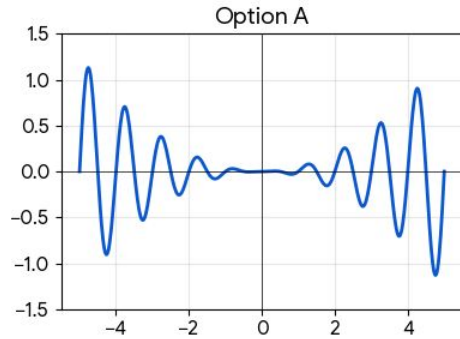
21) Which curve represent the equation $y = e^{(-0.3x)} * \sin(\pi * x)$



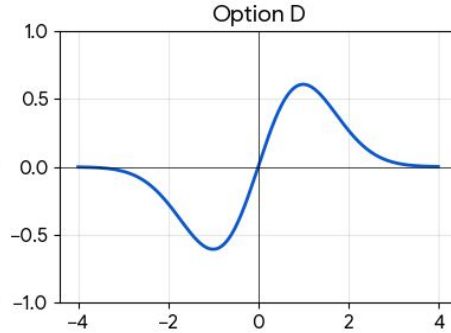
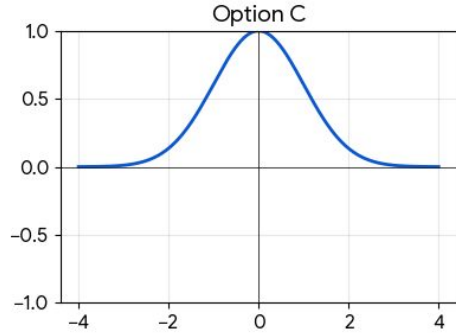
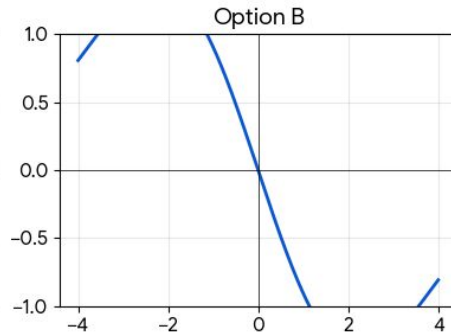
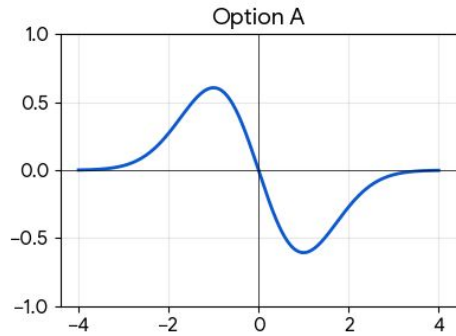
22) Which curve represent the equation $y = \sin(4x) / (1 + x^2)$



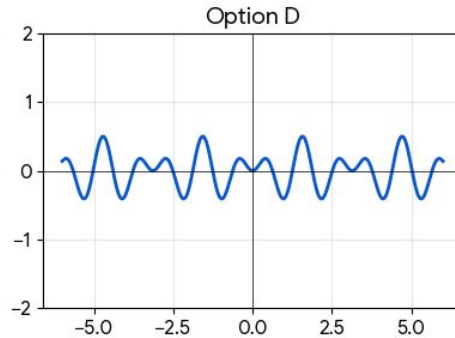
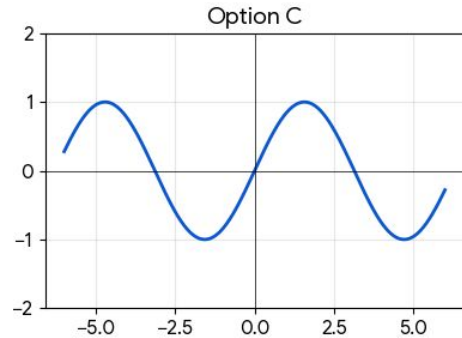
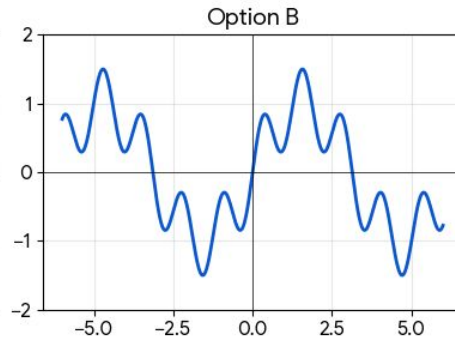
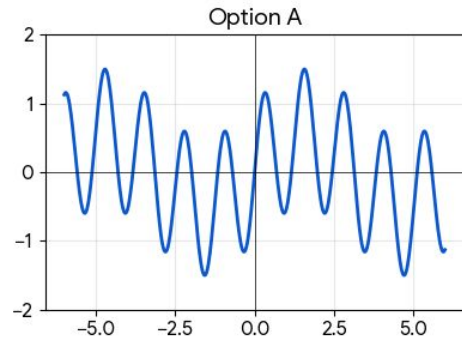
23) Which curve represent the equation $y = 0.2 * x * \sin(2 * \pi * x)$



24) Which curve represent the equation $y = -x * e^{(-0.5 * x^2)}$

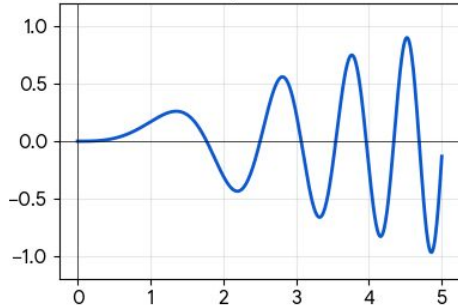


25) Which curve represent the equation $y = \sin(x) + 0.5 * \sin(5 * x)$

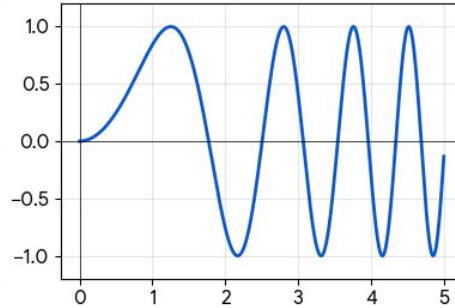


26) Which curve represent the equation $y = \sin(x^2)$

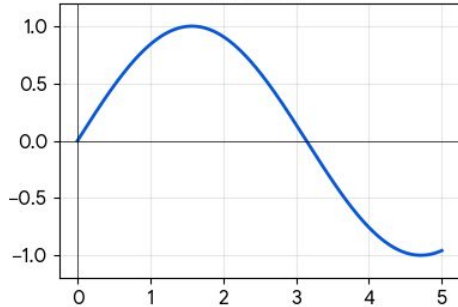
Option A



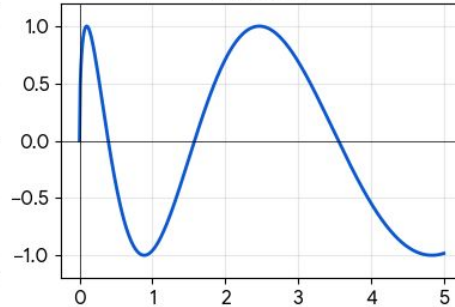
Option B



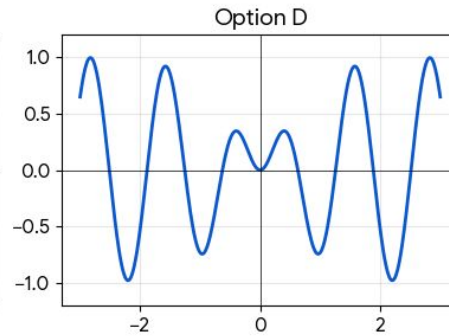
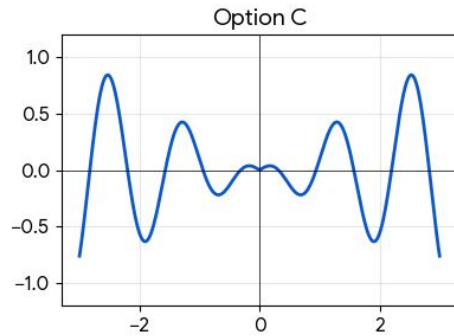
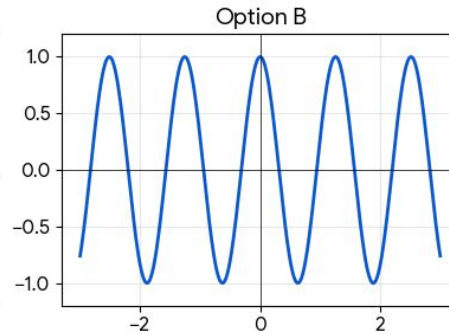
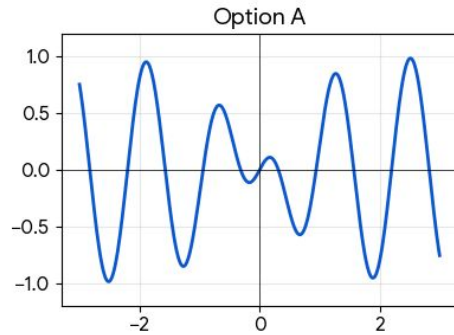
Option C



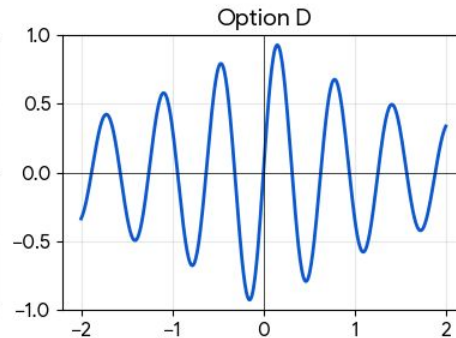
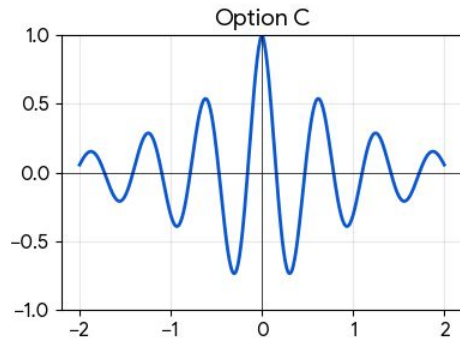
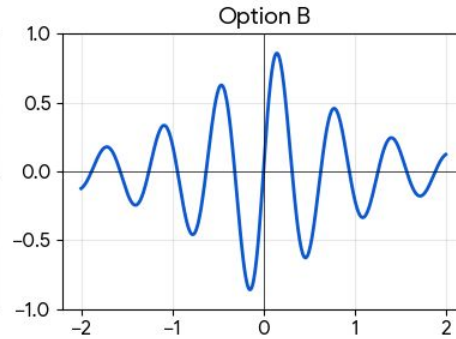
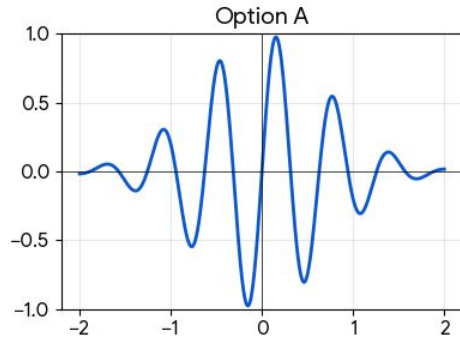
Option D



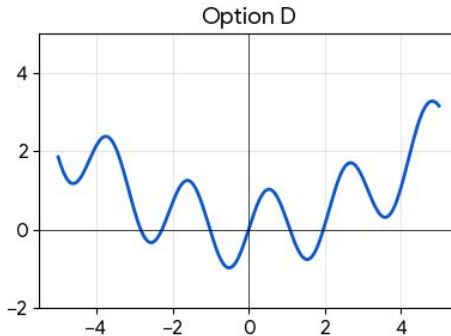
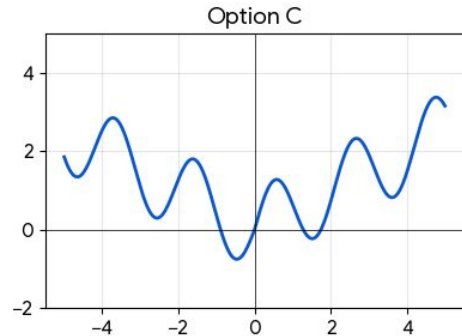
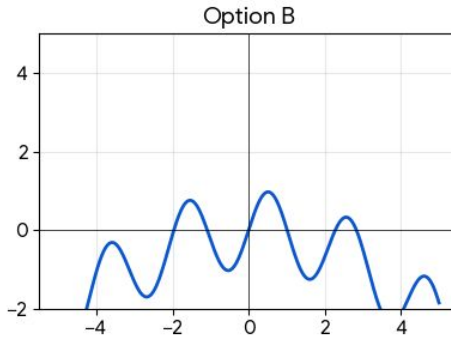
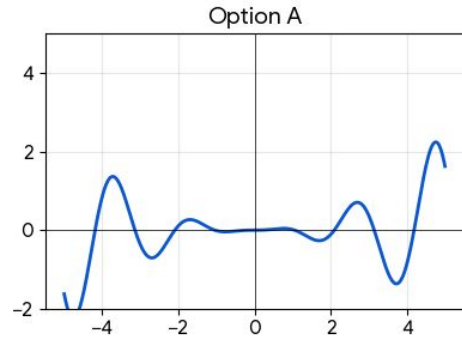
27) Which curve represent the equation $y = \tanh(x) * \cos(5 * x)$



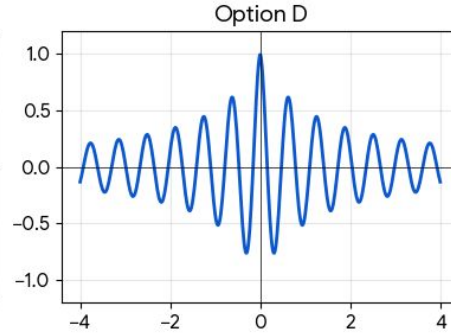
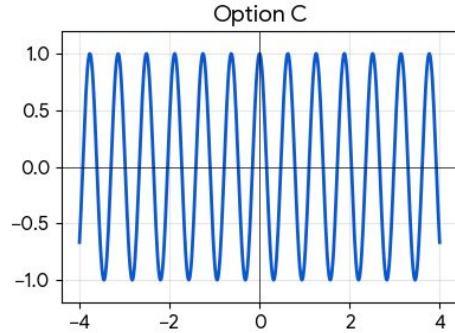
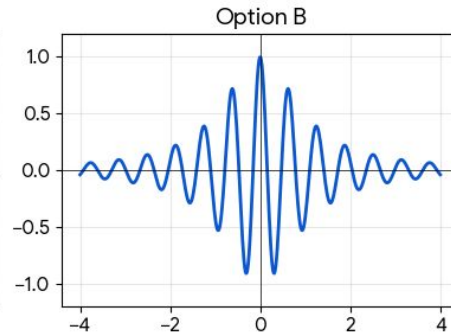
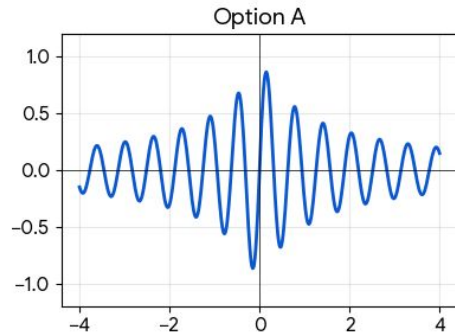
28) Which curve represent the equation $y = e^{-|x|} * \sin(10 * x)$



29) Which curve represent the equation $y = 0.1 * x^2 + \sin(3 * x)$



30) Which curve represent the equation $y = \cos(10 * x) / (1 + |x|)$



Find the domain and range of the following functions:

31) $f(x) = x^3 - 2x^2 + 5$

32) $f(x) = \sqrt{x - 4}$

33) $f(x) = -3 * e^x$

34) $f(x) = \ln(x + 2)$

35) $f(x) = 2 * \sin(x) + 1$

36) $f(x) = 1/x$

37) $f(x) = (x + 1)^{(1/3)}$

38) $f(x) = x^2 - 4$

39) $f(x) = \cos(3x)$

40) $f(x) = -\log(x)$

Extra) For each of these functions, give the asymptotes (if they exist)

41) Find the inverse of $0.5(x + 1)^3 - 2$.

Plot the function and the inverse in the range $x = (-15, 10)$

**42) Find the inverse of $3 * \sqrt{x - 2} + 1$.
Plot the function and the inverse in the range $x = (1, 8)$**

43) Find the inverse of $2 * e^x - 3$.

Plot the function and the inverse in the range $x = (-2, 6)$

44) Find the inverse of $-\ln(x) + 2$.

Plot the function and the inverse in the range $x = (0, 6)$

45) Find the inverse of $3 * \sin(x)$.

Plot the function and the inverse in the range $x = (-\pi/2, \pi/2)$

46) Find the inverse of $(2x + 3) / (x - 1)$.

Plot the function and the inverse in the range $x = (1, 12)$

47) Find the inverse of $(x - 4)^{1/3} + 2$.

Plot the function and the inverse in the range $x = (-5, 10)$

**48) Find the inverse of $2(x - 3)^2 + 1$, ($x > 3$).
Plot the function and the inverse in the range $x = (1, 10)$**

49) Find the inverse of $\cos(2x)$.

Plot the function and the inverse in the range $x = (0, \pi/2)$

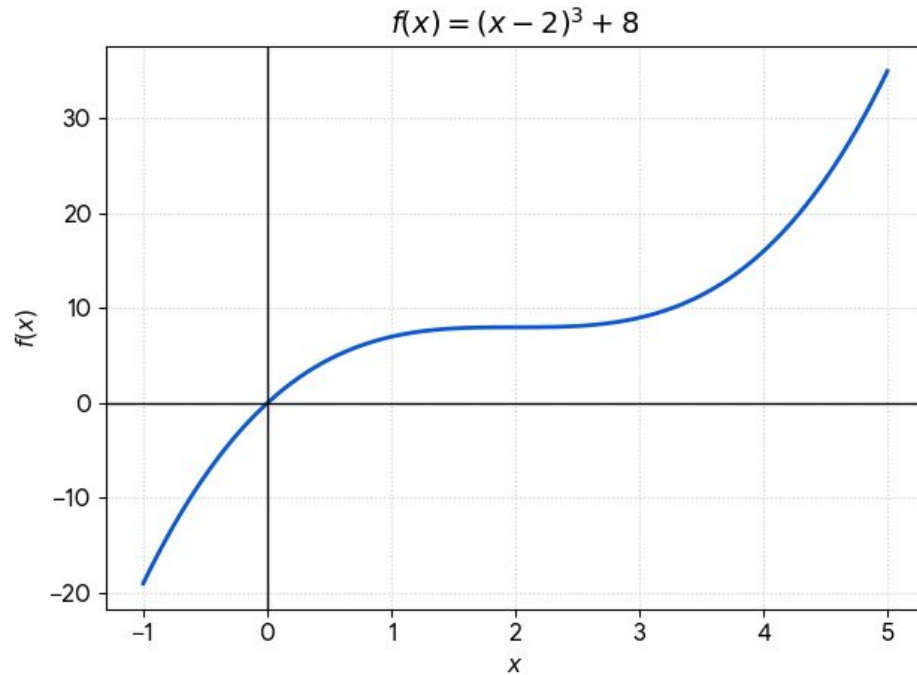
50) Find the inverse of $\log_{10}(2x + 10)$.

Plot the function and the inverse in the range $x = (-4, 10)$

SOLUTIONS

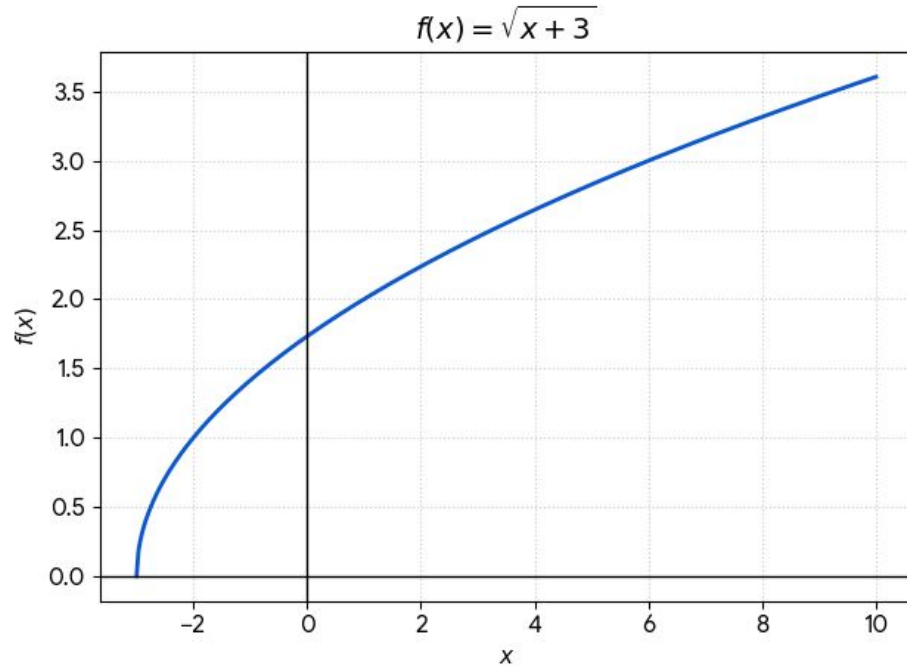
1) Which function does this curve represent?

- A. $f(x) = (x+2)^3$
- B. $f(x) = x^3 - 2$
- C. $f(x) = (x-2)^3 + 8$
- D. $f(x) = 1/2(x-2)^3$



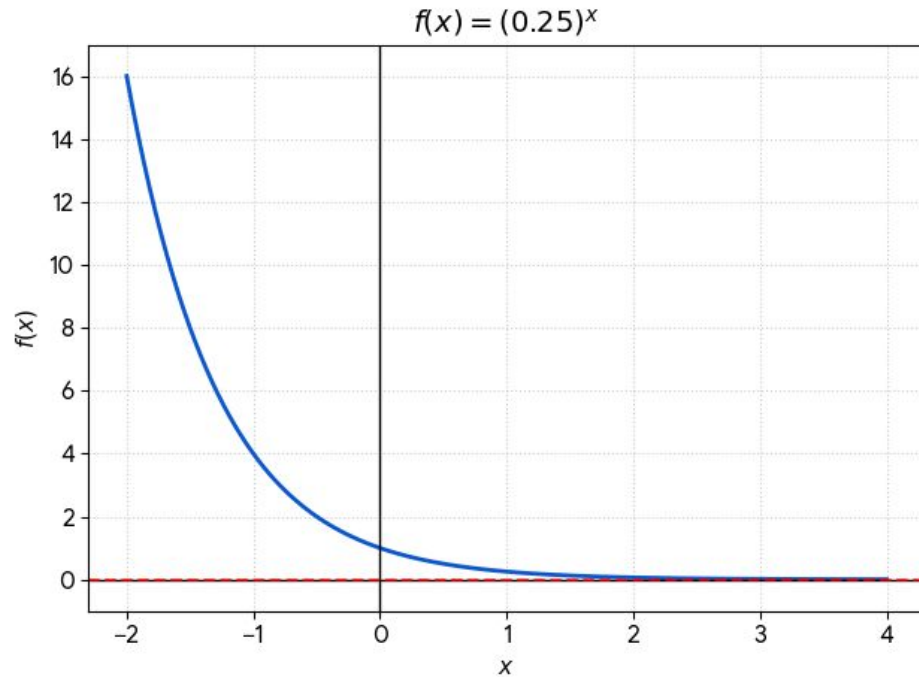
2) Which function does this curve represent?

- A. $f(x) = \sqrt{x} - 3$
- B. $f(x) = \sqrt{x+3}$
- C. $f(x) = \sqrt{x-3}$
- D. $f(x) = 2\sqrt{x+3}$



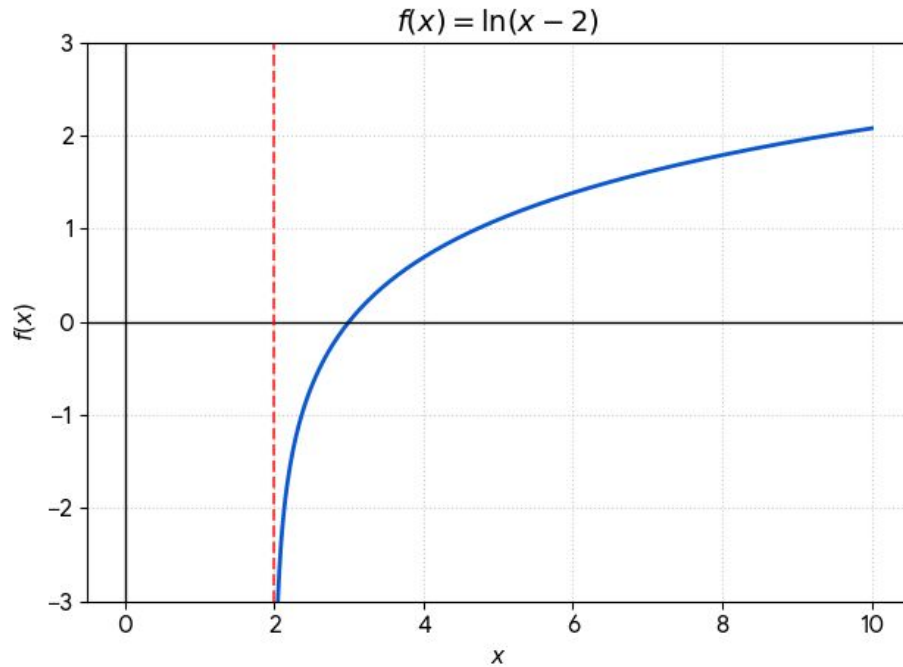
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- A. $f(x) = 4^x$
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- C. $f(x) = (0.25)^x$
- D. $f(x) = -4^x$



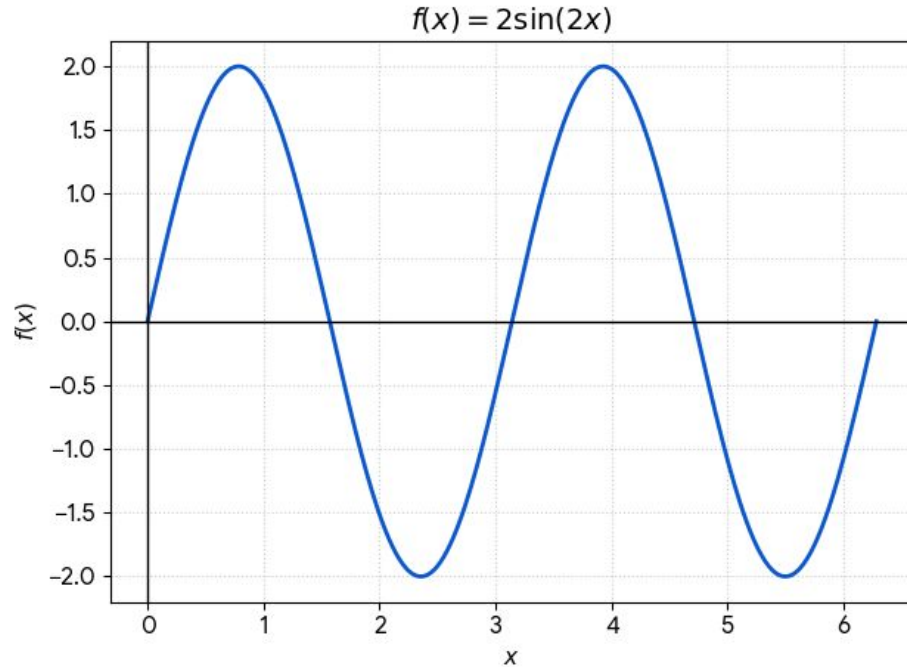
4) Which function does this curve represent?

- A. $f(x) = \ln(x) + 2$
- B. $f(x) = \ln(x+2)$
- C. $f(x) = \ln(x-2)$
- D. $f(x) = 2 \cdot \ln(x)$



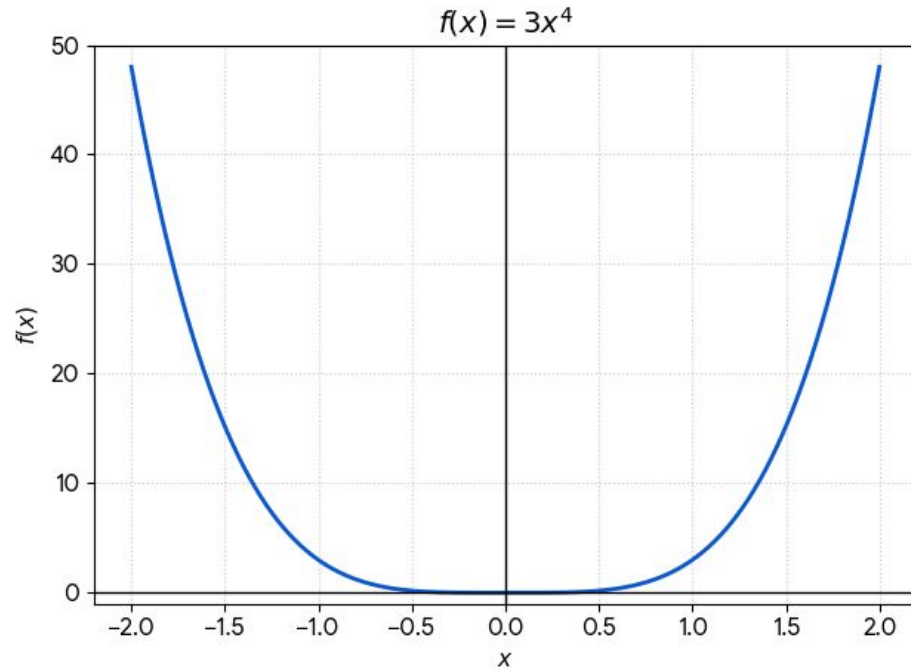
5) Which function does this curve represent?

- A. $f(x) = 2\sin(x)$
- B. $f(x) = \sin(2x)$
- C. $f(x) = 2\sin(2x)$
- D. $f(x) = 1/2\sin(\pi x)$



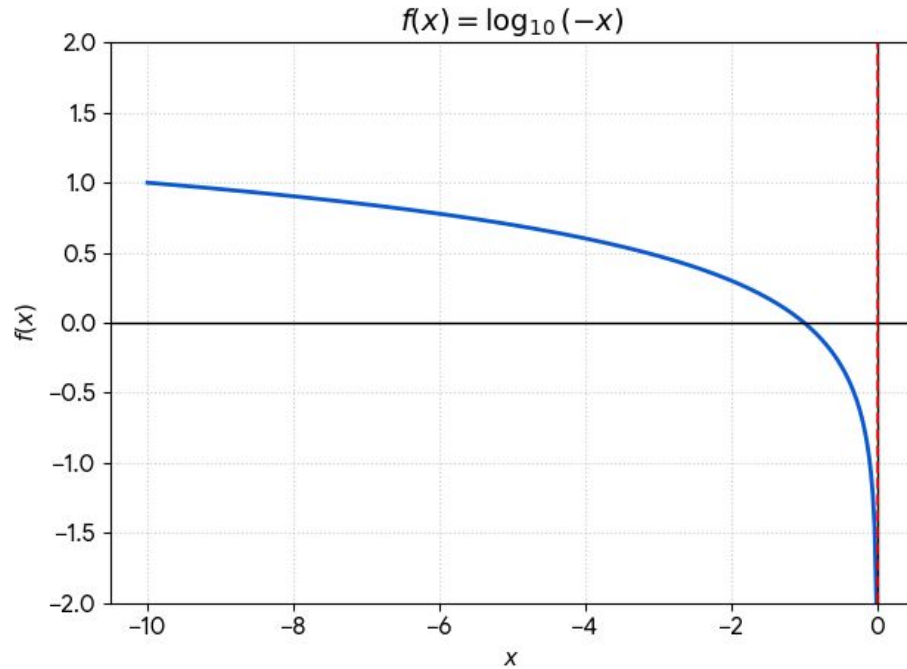
6) Which function does this curve represent?

- A. $f(x) = 3x^2$
- B. $f(x) = x^4 + 2$
- C. $f(x) = 3x^4$
- D. $f(x) = x^6 + 3$



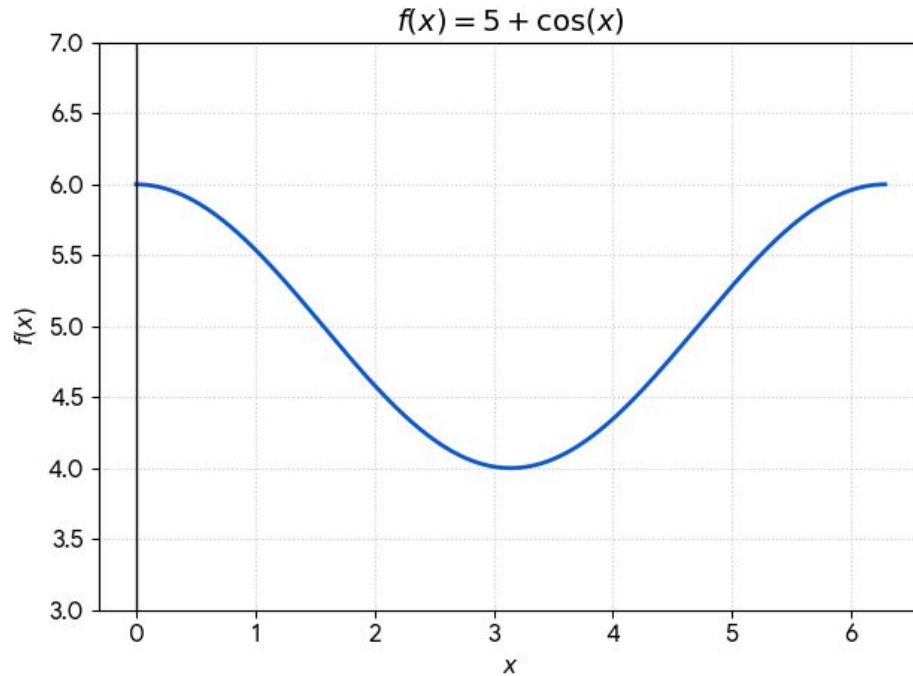
7) Which function does this curve represent?

- A. $f(x) = \log_{10}(x)$
- B. $f(x) = -\log_{10}(x)$
- C. $f(x) = \log_{10}(-x)$
- D. $f(x) = \log_{10}(x-10)$



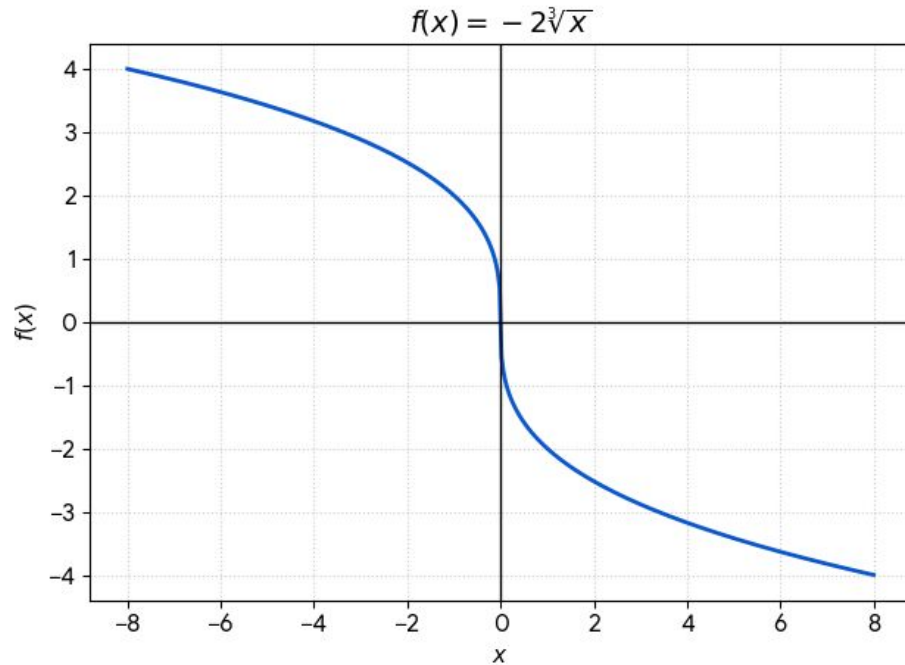
8) Which function does this curve represent?

- A. $f(x) = 5 + \cos(x)$
- B. $f(x) = 6\cos(x)$
- C. $f(x) = \cos(x) + 4$
- D. $f(x) = 2\cos(x) + 4$



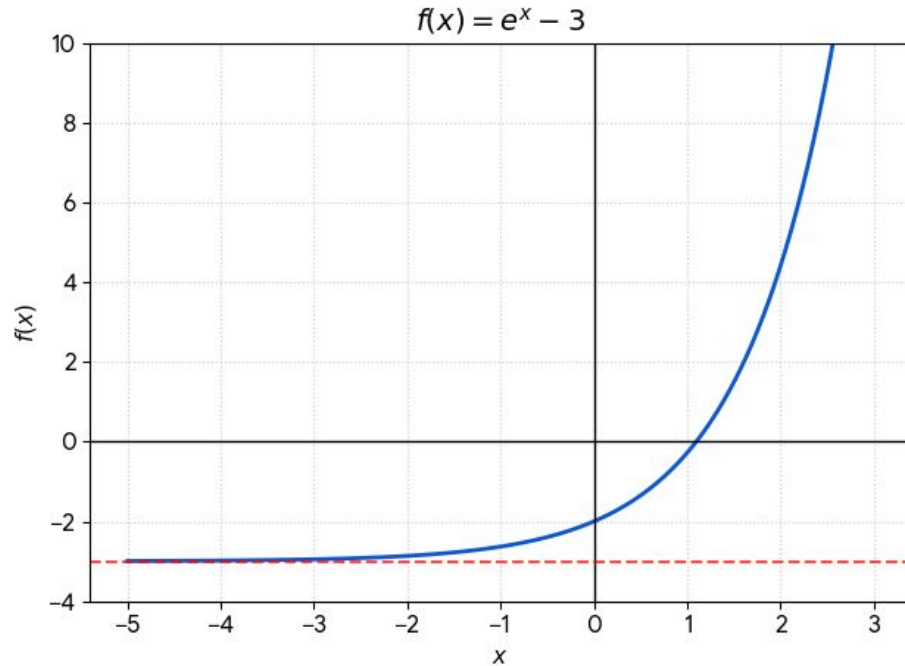
9) Which function does this curve represent?

- A. $f(x) = -2 \cdot \text{cbrt}(x)$
- B. $f(x) = \text{cbrt}(x-2)$
- C. $f(x) = \text{cbrt}(-2x)$
- D. $f(x) = \text{cbrt}(x) - 2$



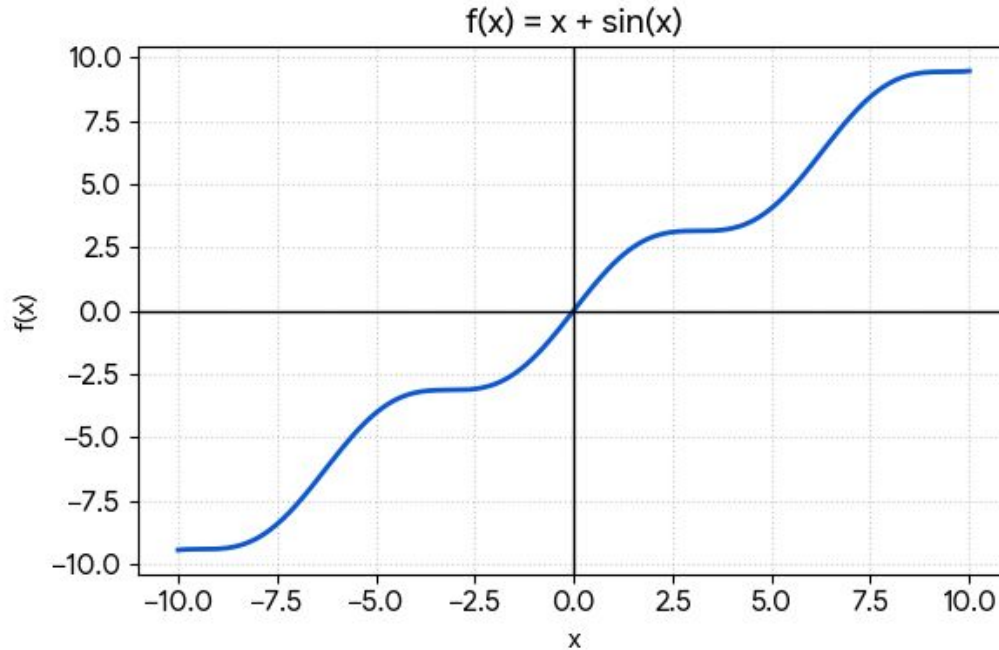
10) Which function does this curve represent?

- A. $f(x) = e^x + 3$
- B. $f(x) = e^{(x+3)}$
- C. $f(x) = e^x - 3$
- D. $f(x) = 3 \cdot e^x - 3$



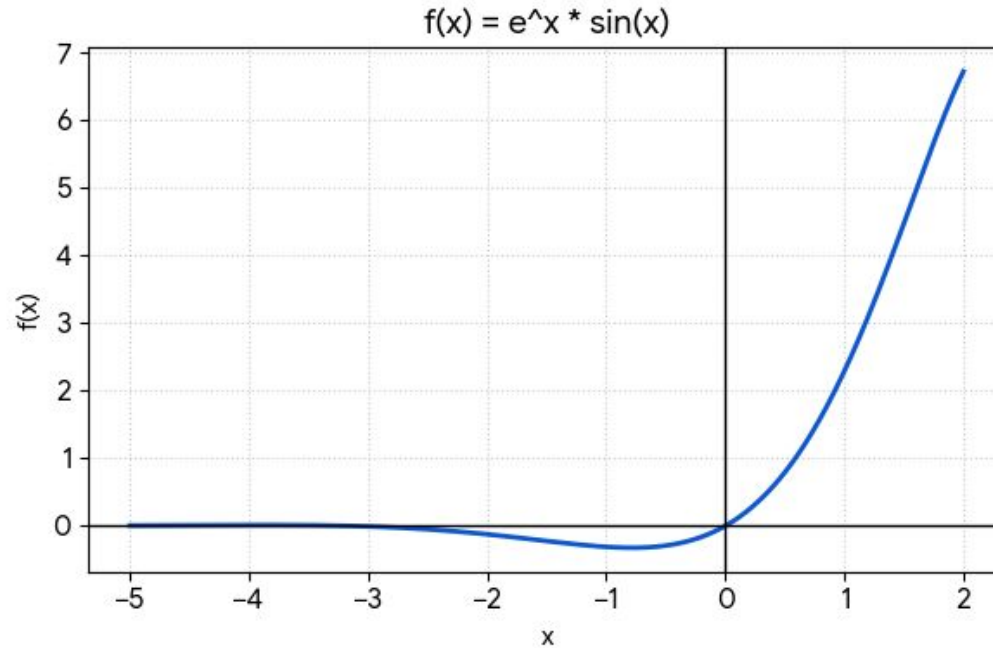
11) Which function does this curve represent?

- A. $f(x) = x + \sin(x)$
- B. $f(x) = x * \sin(x)$
- C. $f(x) = x + \cos(x)$
- D. $f(x) = x^2 + \sin(x)$



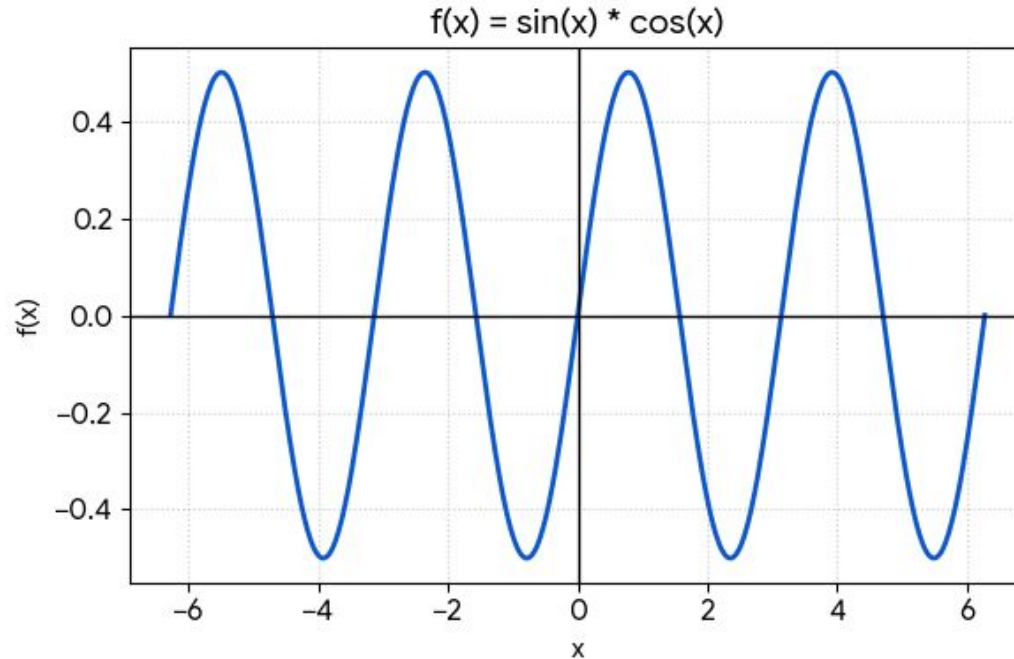
12) Which function does this curve represent?

- A. $f(x) = e^x + \sin(x)$
- B. $f(x) = e^x * \sin(x)$
- C. $f(x) = e^{-x} * \sin(x)$
- D. $f(x) = \ln(x) * \sin(x)$



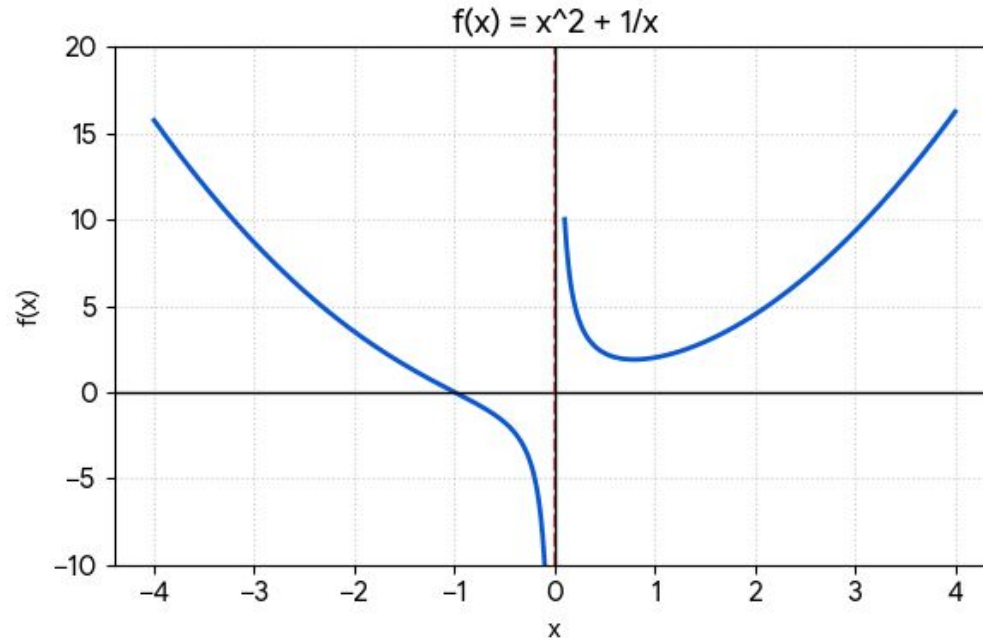
13) Which function does this curve represent?

- A. $f(x) = \sin(x) + \cos(x)$
- B. $f(x) = \sin(x) * \cos(x)$
- C. $f(x) = \sin(x) / \cos(x)$
- D. $f(x) = \sin(x) + x^2$



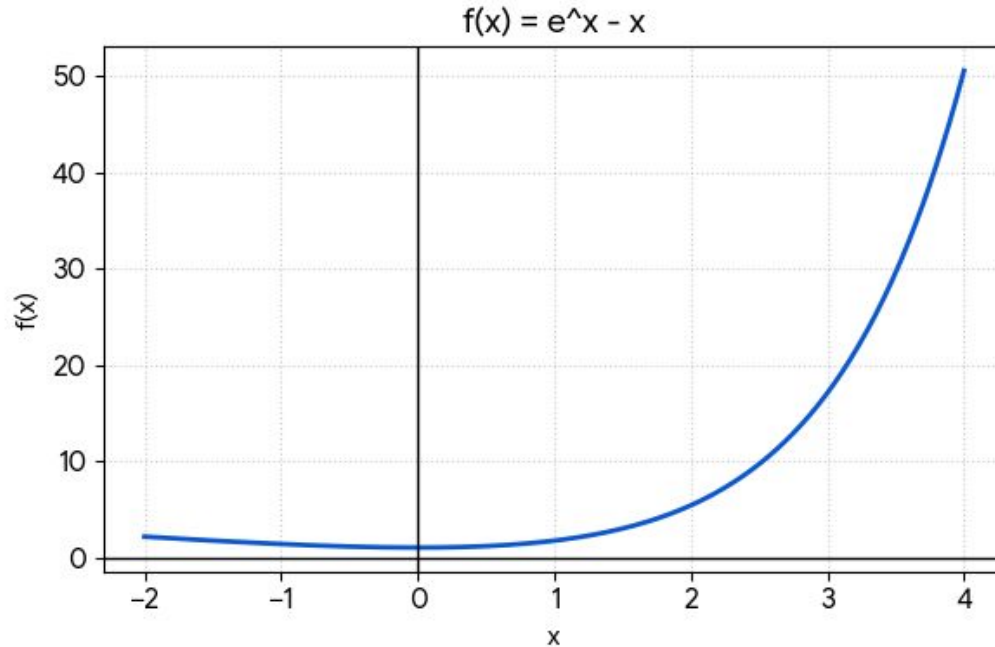
14) Which function does this curve represent?

- A. $f(x) = x^2 + \ln(x)$
- B. $f(x) = x^2 - e^x$
- C. $f(x) = x^2 + 1/x$
- D. $f(x) = x^2 * \ln(x)$



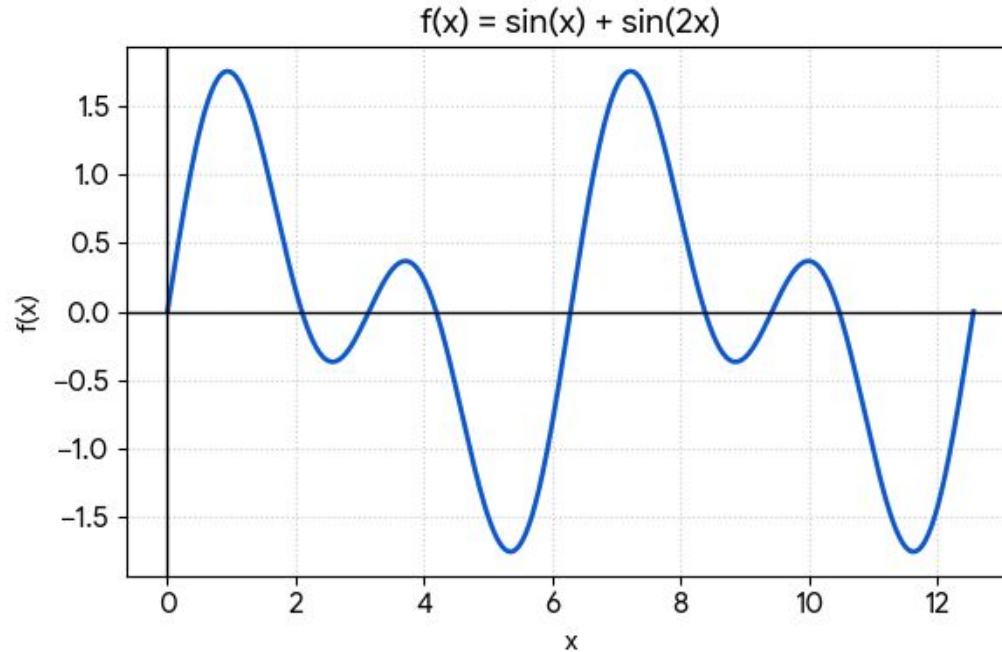
15) Which function does this curve represent?

- A. $f(x) = e^x + x$
- B. $f(x) = e^x - x$
- C. $f(x) = e^{-x} + x$
- D. $f(x) = e^x * x$



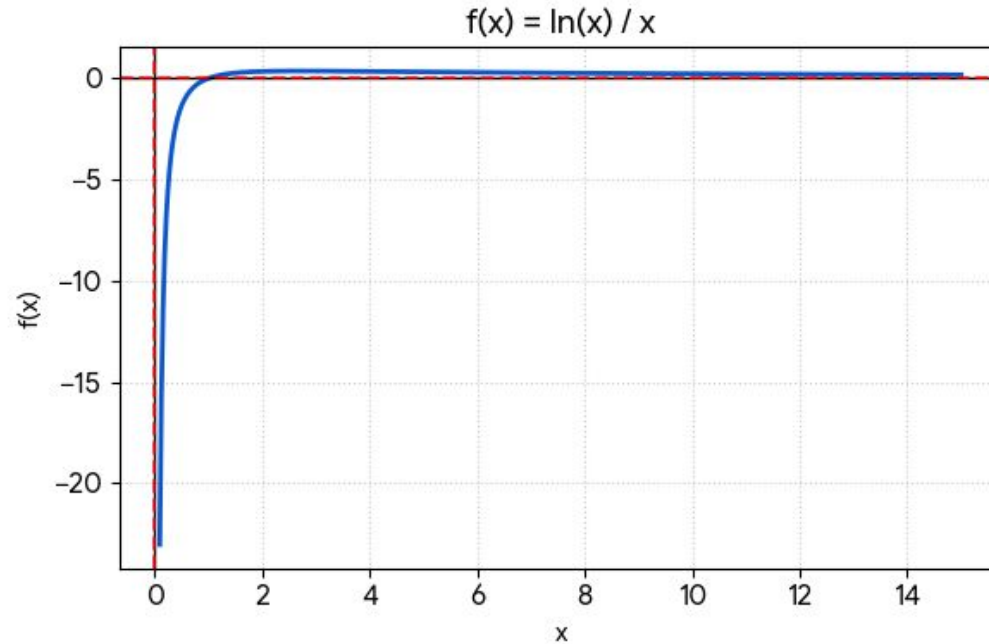
16) Which function does this curve represent?

- A. $f(x) = \sin(x) + \sin(2x)$
- B. $f(x) = \sin(x) + \cos(x)$
- C. $f(x) = \sin(x) * \sin(2x)$
- D. $f(x) = \text{abs}(\sin(x))$



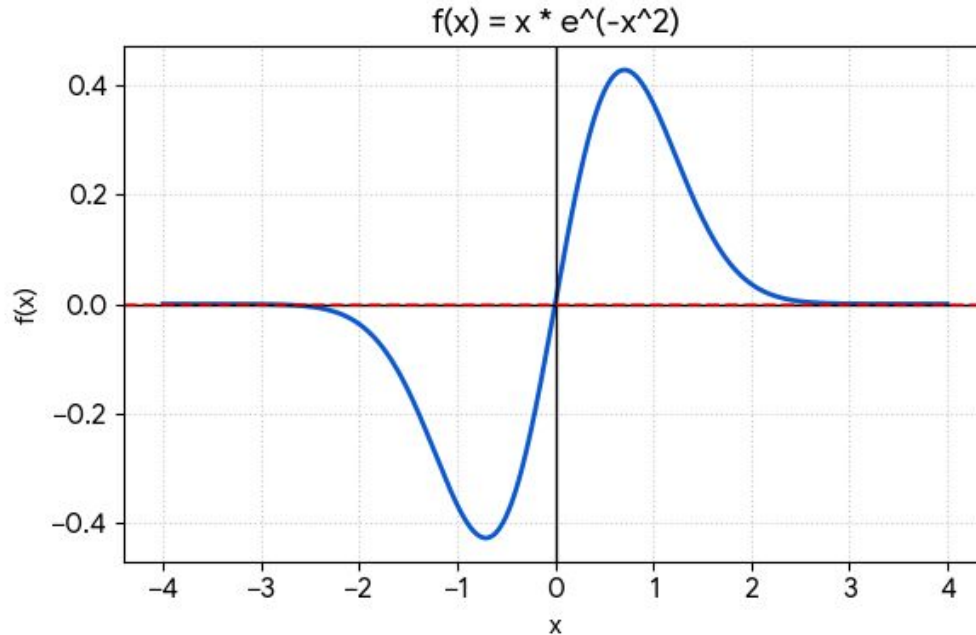
17) Which function does this curve represent?

- A. $f(x) = \ln(x) + x$
- B. $f(x) = \ln(x) - x$
- C. $f(x) = \ln(x) / x$
- D. $f(x) = \ln(x^2)$



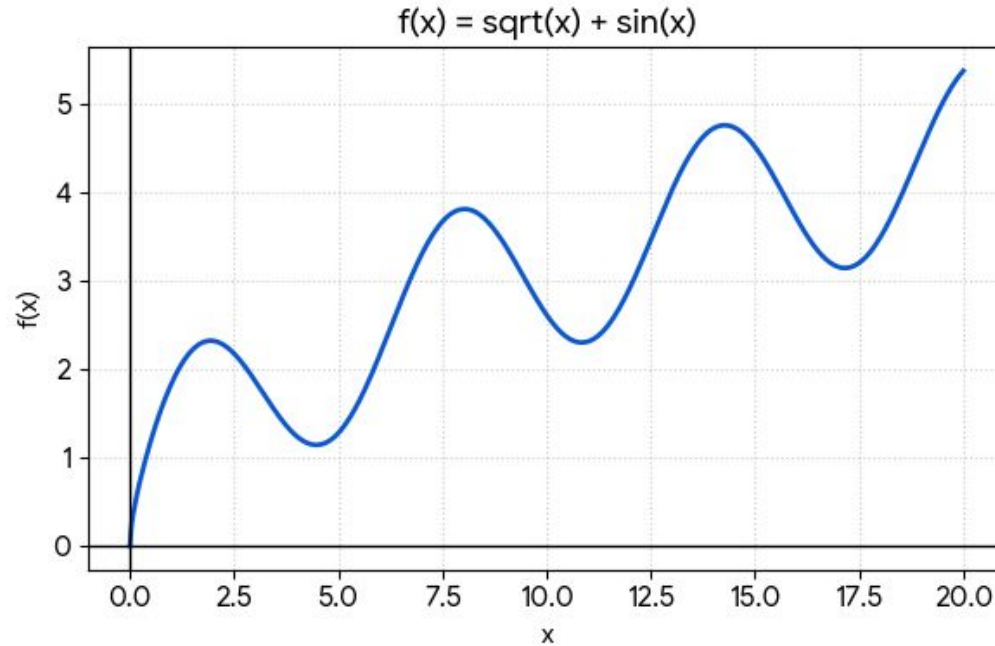
18) Which function does this curve represent?

- A. $f(x) = x * e^{(-x^2)}$
- B. $f(x) = x + e^{(-x)}$
- C. $f(x) = x^2 * e^x$
- D. $f(x) = e^{(-x)} * \cos(x)$



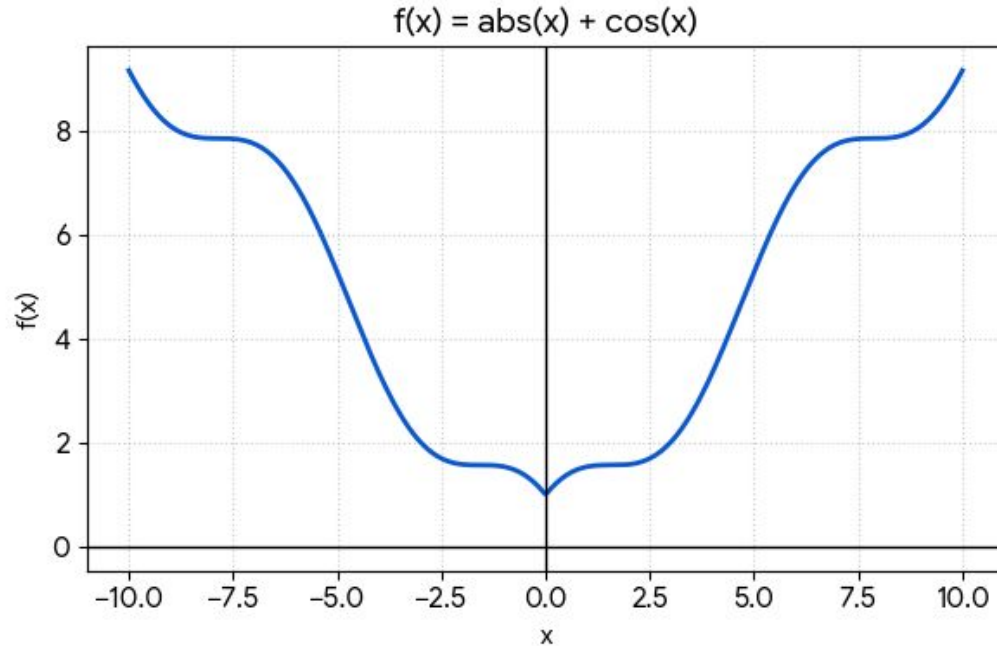
19) Which function does this curve represent?

- A. $f(x) = \sqrt{x} + \sin(x)$
- B. $f(x) = \sqrt{x} * \cos(x)$
- C. $f(x) = 1/x + \sin(x)$
- D. $f(x) = x^{1/3} + x$

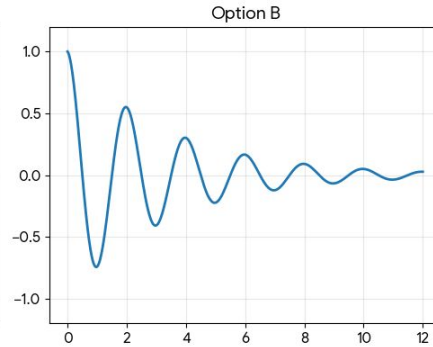
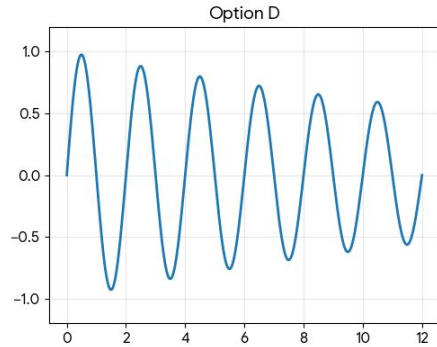
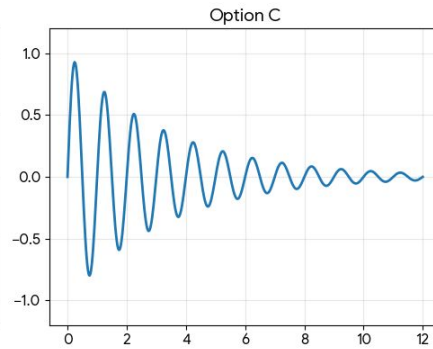
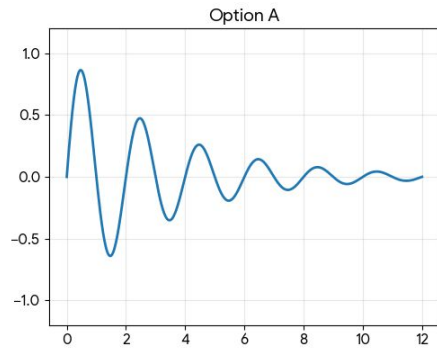


20) Which function does this curve represent?

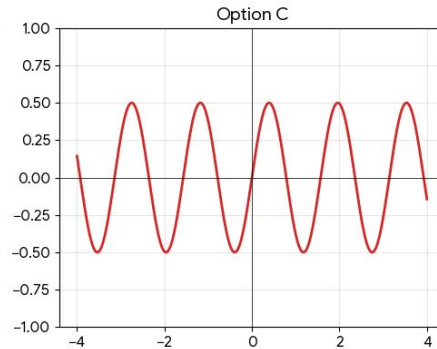
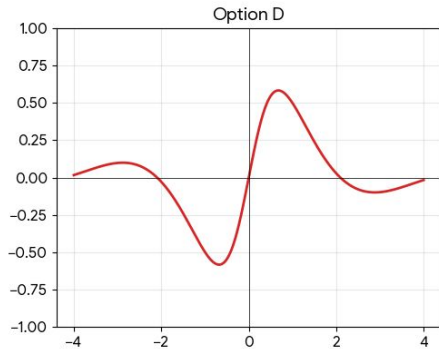
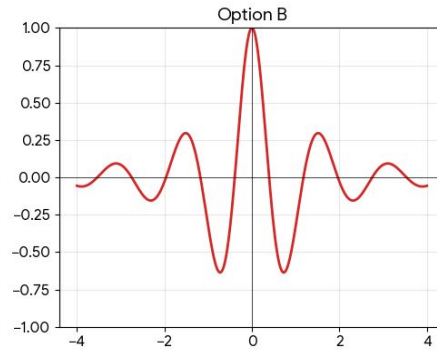
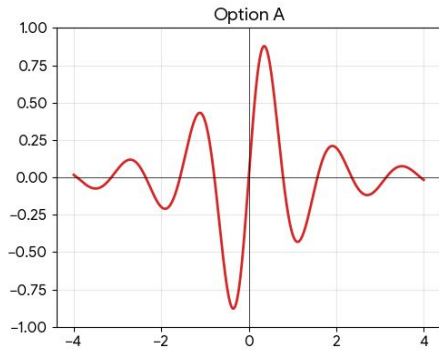
- A. $f(x) = \text{abs}(x) + \cos(x)$
- B. $f(x) = x^2 + \cos(x)$
- C. $f(x) = \text{abs}(x) * \sin(x)$
- D. $f(x) = 1/x + \cos(x)$



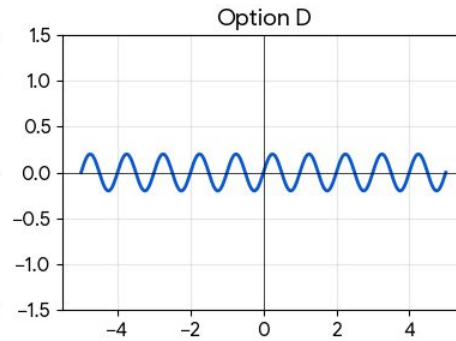
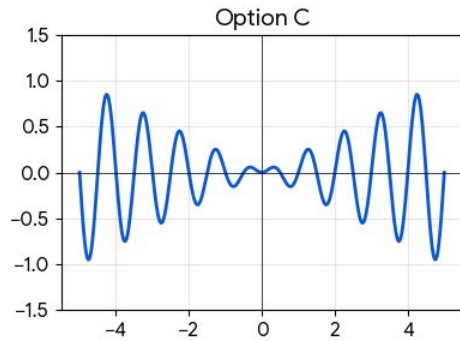
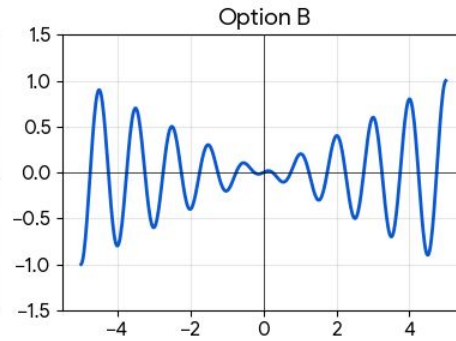
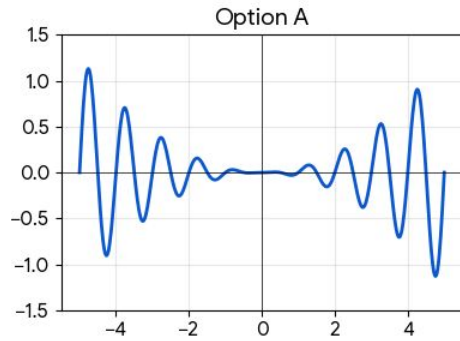
21) Option A



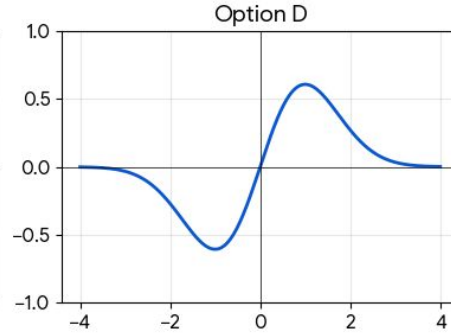
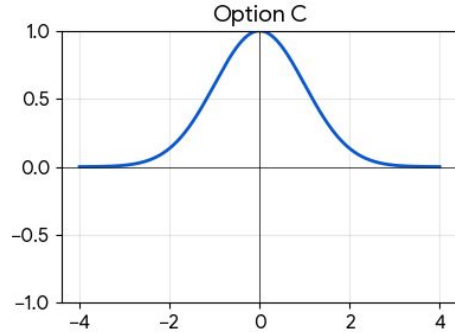
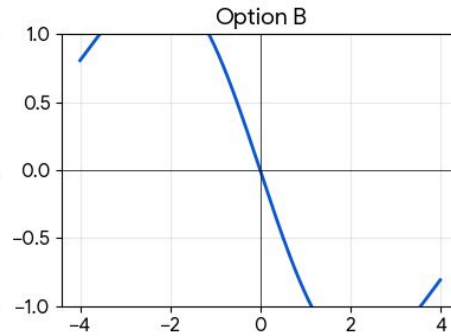
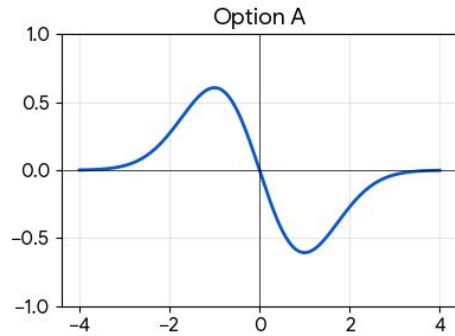
22) Option A



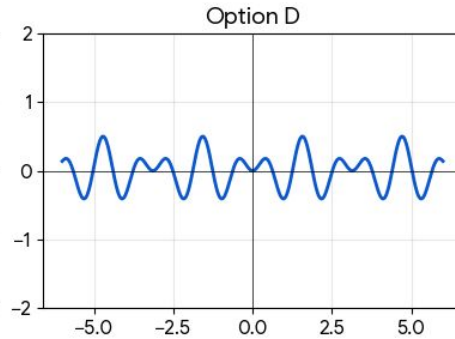
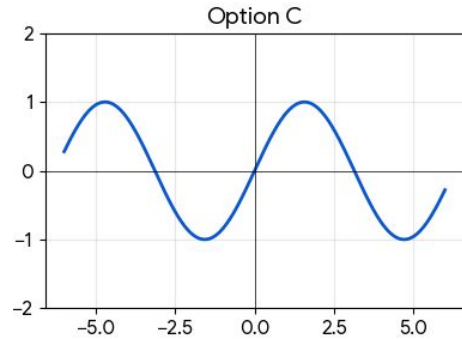
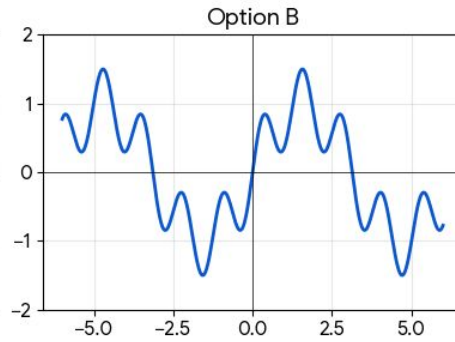
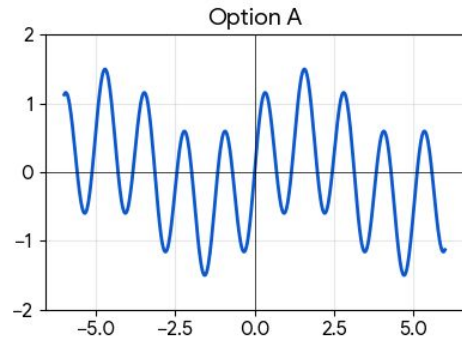
23) Option C



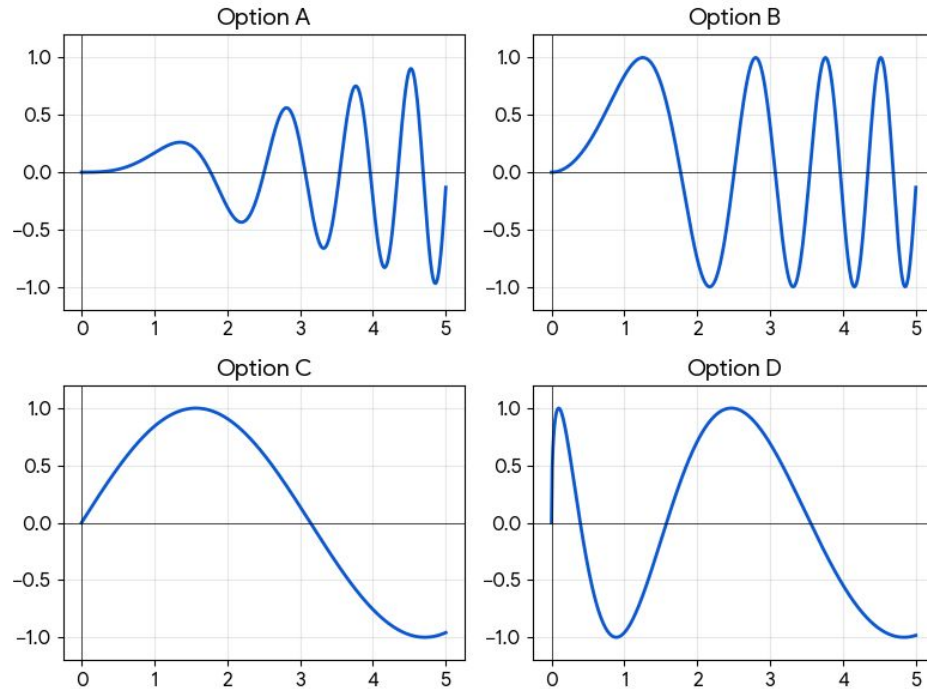
24) Option A



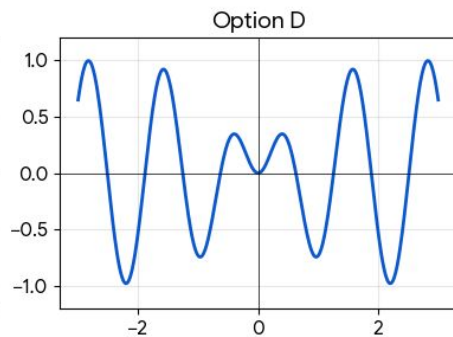
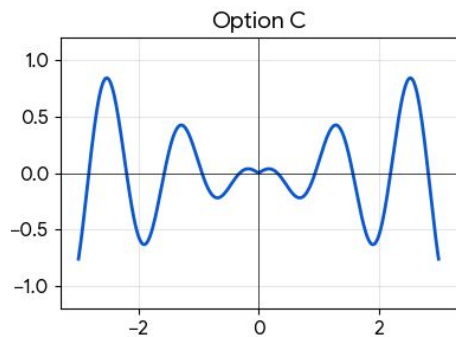
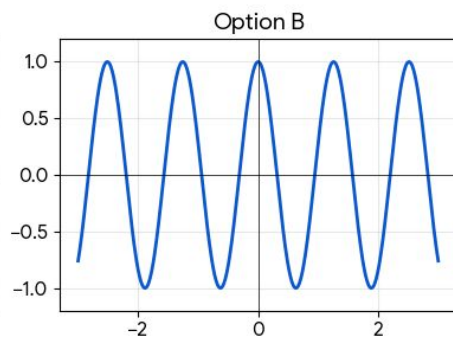
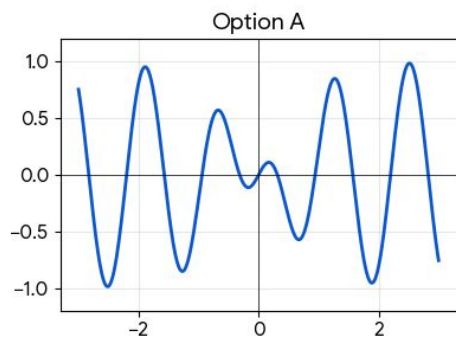
25) Option B



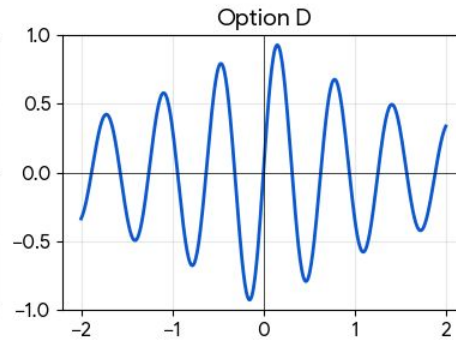
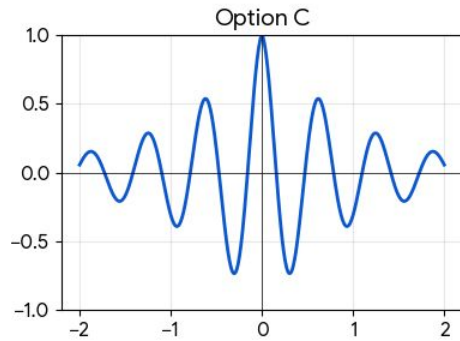
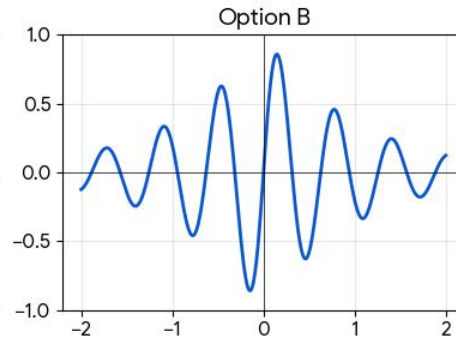
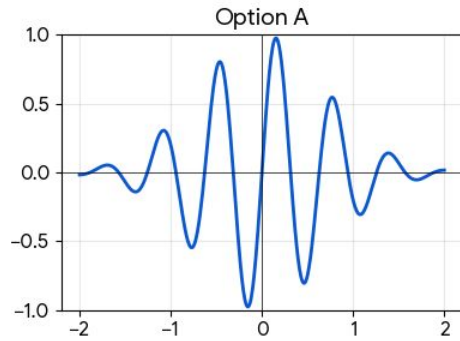
26) Option B



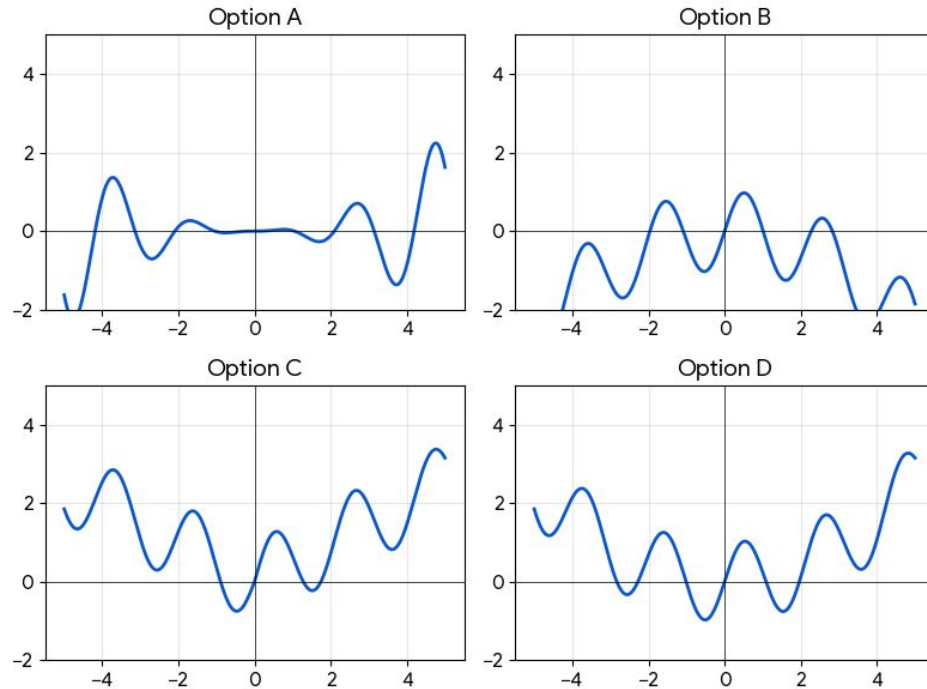
27) Option A



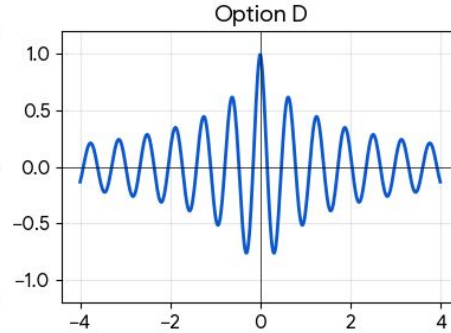
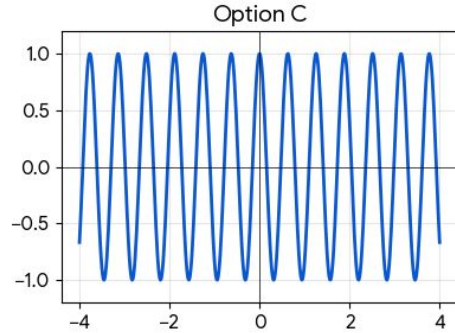
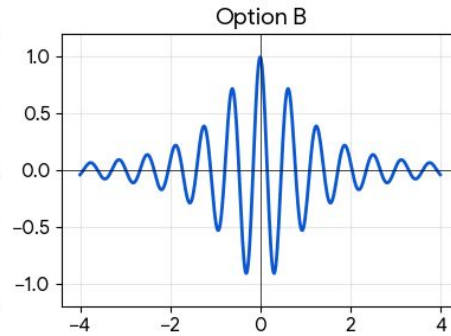
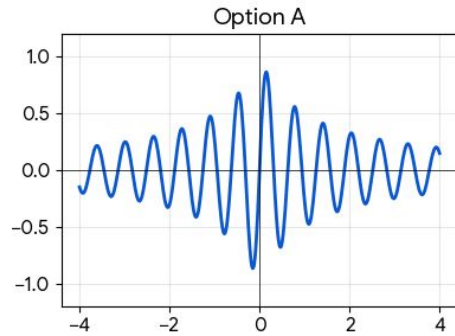
28) Option B



29) Option D



30) Option D



Find the domain and range of the following functions:

- 31) Domain: $(-\infty, \infty)$, Range: $(-\infty, \infty)$
- 32) Domain: $[4, \infty)$, Range: $[0, \infty)$
- 33) Domain: $(-\infty, \infty)$, Range: $(-\infty, 0)$
- 34) Domain: $(-2, \infty)$, Range: $(-\infty, \infty)$
- 35) Domain: $(-\infty, \infty)$, Range: $[-1, 3]$
- 36) Domain: $(-\infty, 0) \cup (0, \infty)$, Range: $(-\infty, 0) \cup (0, \infty)$
- 37) Domain: $(-\infty, \infty)$, Range: $(-\infty, \infty)$
- 38) Domain: $(-\infty, \infty)$, Range: $[-4, \infty)$
- 39) Domain: $(-\infty, \infty)$, Range: $[-1, 1]$
- 40) Domain: $(0, \infty)$, Range: $(-\infty, \infty)$

Extra)

31) No asymptote

32) No asymptote

33) Horizontal Asymptote at $y = 0$

34) Vertical Asymptote at $x = -2$

35) No asymptote

36) Vertical Asymptote at $x = 0$, Horizontal Asymptote at $y = 0$

37) No asymptote

38) No asymptote

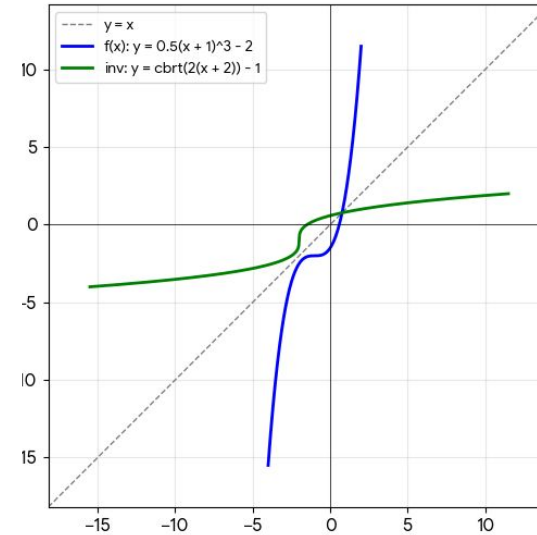
39) No asymptote

40) Vertical Asymptote at $x = 0$

41) Find the inverse of $0.5(x + 1)^3 - 2$.

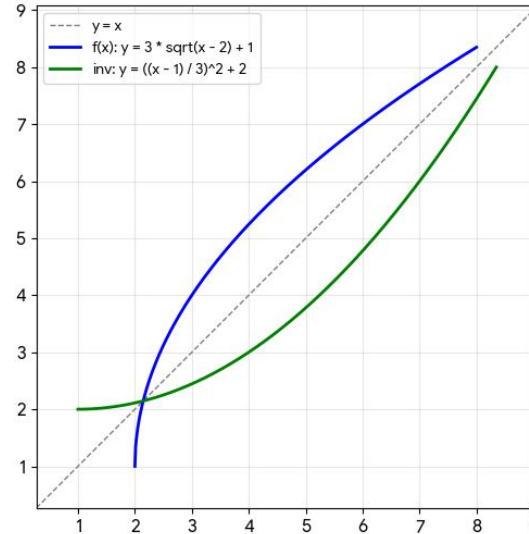
Plot the function and the inverse in the range $x = (-15, 10)$

$$f^{-1}(x) = (2(x + 2))^{1/3} - 1$$



**42) Find the inverse of $3 * \sqrt{x - 2} + 1$.
Plot the function and the inverse in the range $x = (1, 8)$**

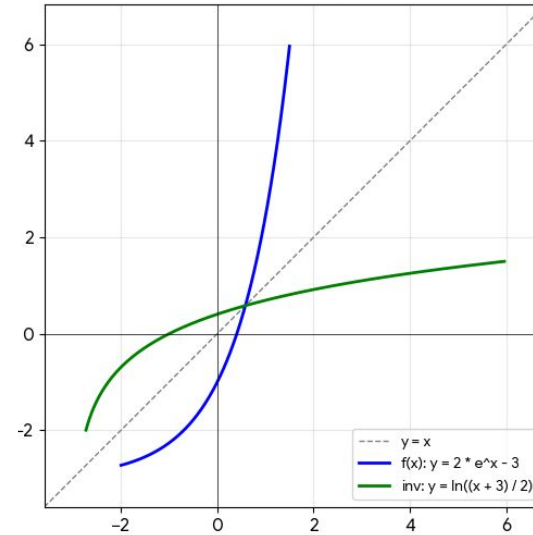
$$f^{-1}(x) = ((x - 1) / 3)^2 + 2$$



43) Find the inverse of $2 * e^x - 3$.

Plot the function and the inverse in the range $x = (-2, 6)$

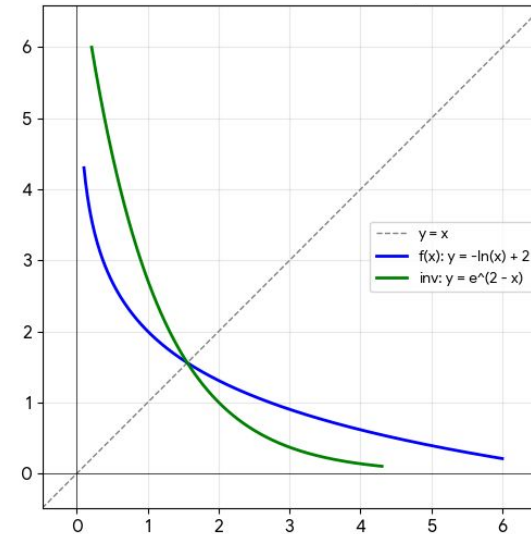
$$f^{-1}(x) = \ln((x + 3) / 2)$$



44) Find the inverse of $-\ln(x) + 2$.

Plot the function and the inverse in the range $x = (0, 6)$

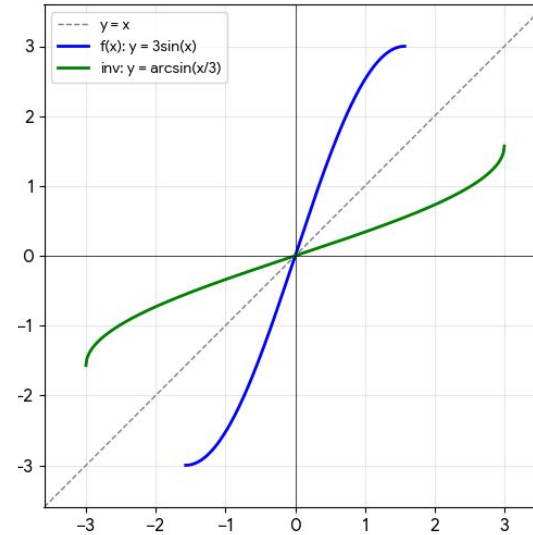
$$f^{-1}(x) = e^{2-x}$$



45) Find the inverse of $3 \cdot \sin(x)$.

Plot the function and the inverse in the range $x = (-\pi/2, \pi/2)$

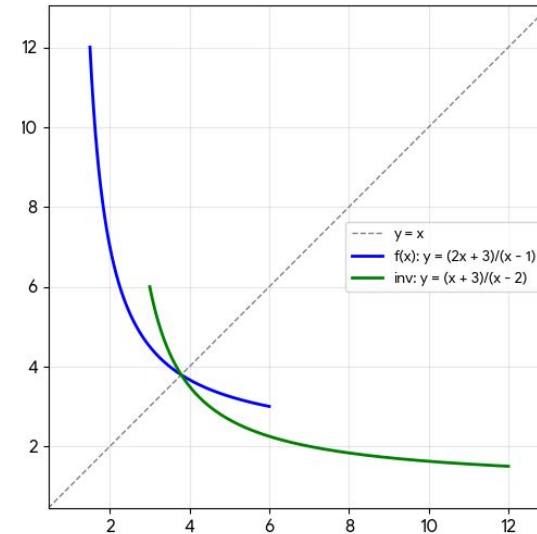
$$f^{-1}(x) = \arcsin(x / 3)$$



46) Find the inverse of $(2x + 3) / (x - 1)$.

Plot the function and the inverse in the range $x = (1, 12)$

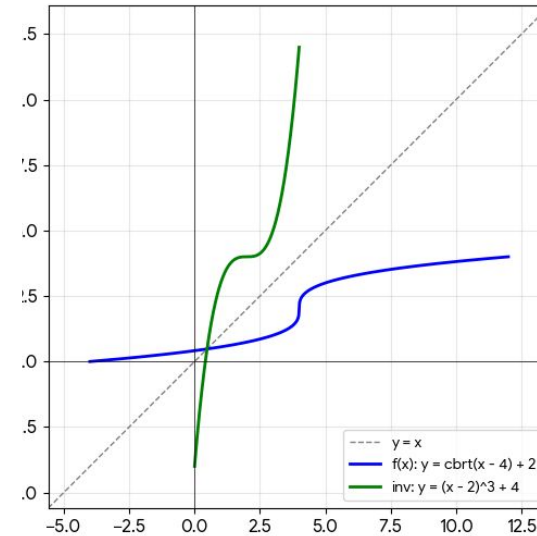
$$f^{-1}(x) = (x + 3) / (x - 2)$$



47) Find the inverse of $(x - 4)^{1/3} + 2$.

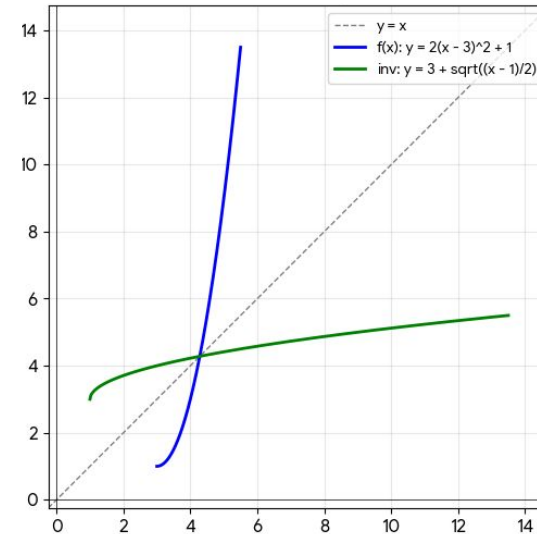
Plot the function and the inverse in the range $x = (-5, 10)$

$$f^{-1}(x) = (x - 2)^3 + 4$$



**48) Find the inverse of $2(x - 3)^2 + 1$, ($x > 3$).
Plot the function and the inverse in the range $x = (1, 10)$**

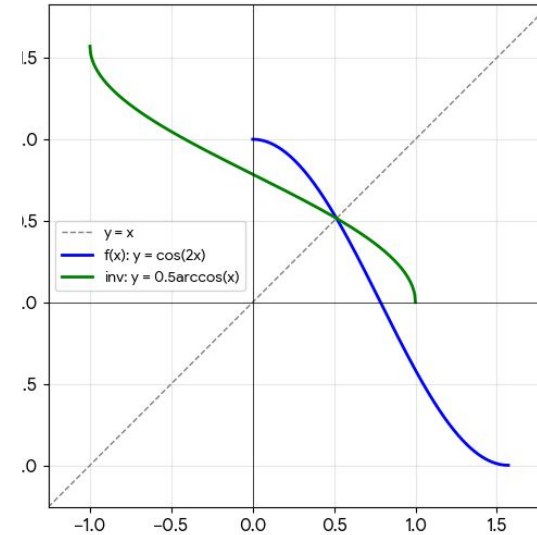
$$f^{-1}(x) = 3 + \sqrt{(x - 1) / 2}$$



49) Find the inverse of $\cos(2x)$.

Plot the function and the inverse in the range $x = (0, \pi/2)$

$$f^{-1}(x) = 0.5 * \arccos(x)$$



50) Find the inverse of $\log_{10}(2x + 10)$.

Plot the function and the inverse in the range $x = (-4, 10)$

$$f^{-1}(x) = (10^x / 2) - 5$$

